

Sub Code: AHT-006 ROLL NO..... EVEN SEMESTER EXAMINATION,
2023 – 24

2nd year B.Tech.

Advanced Applied mathematics

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. $\infty - 3???$ (a) Evaluate by Laplace transform $\int ?????$ $???, ????.$ b) Verify final value theorem for $??(??) = 1 + ??^{-??}(???? + ????).$ c) Find the Laplace transform of the periodic function: $f(t) = \{????\}$ for $0 < ?? < ??$ $0 < ?? < 2??.$ d) State convolution theorem and hence find $??^{-1}(^{??2}$ $(??^2 + ??^2)(??^2 + ??^2)).$ e) Using unit step function, find the Laplace transform of : (i) $(?? - 1)^{2??}(?? - 1)$ (ii) $\sin t u(t - \pi).$ f) Using Laplace transform method , solve $(tD^2 + (1-2t)D - 2)y=0$, if $y(0)=1, y'(0)=2$. where $D = ^{??}???$	5x4=20
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Q 2.	Answer any four parts of the following. a) Find the Fourier transform of : $f(x) = \begin{cases} 1, & x < 1 \\ 0, & x > 1 \end{cases}$ Hence evaluate $\int_{-\infty}^{\infty} f(x) \sin x \, dx$. b) Using Fourier sine integral of the function $f(x) = e^{-ax}$, $a > 0$, show that $\int_0^{\infty} \frac{\sin x}{x} \, dx = \frac{\pi}{2}$. c) Solve the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, under the conditions (i) $u(0, y) = 0$ (ii) $u(x, 0) = e^{-ax}$ (iii) $u(x, y)$ is bounded. d) Find the Fourier cosine transform of the function $f(x)$ if $f(x) = \begin{cases} x, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$. e) Find the Fourier sine transform of $f(x) = e^{-ax}$. Hence find Fourier sine transform of $\frac{1}{x}$. f) Find the inverse Fourier transform of $F(s) = \frac{1}{s^2 + 1}$, where $s \in [-\infty, \infty]$.	5x4=20
Q 3.	Answer any two parts of the following. a) Find the root of the equation $x^3 - 5x^2 - 7 = 0$ which lies between 2 and 3 by the method of false position. b) Using Lagrange's interpolation formula, find $f(8)$ from the following table:	10x2=20

Q 4.

Q 5.

x	4	5	7	10	11	13
f(<u>22</u>)	48	100	294	900	1210	2028

$$10 \times 2 = 20$$

$$10 \times 2 = 20$$

c)The following table gives the population of a town during the last six censuses. Estimate using any suitable interpolation formula, the increase in the population during the period from 1946 to 1948:

Year	1911	1921	1931	1941	1951	1961
Population (in thousands)	12	15	20	27	39	52

Answer any two parts of the following.

a) Find an approximate value of the $\int_0^5 \frac{1}{x^2+5} dx$ by calculating to four decimal places by 5

Simpson's $\frac{1}{3}$ rd rule, $\int_0^5$

by dividing the range into 10 equal parts.

4.2.2

b) Using Runge – Kutta fourth order method, find y for $x = 0.1, 0.2, 0.3$, given that $y(0) = 1$.

Continue the solution at $x = 0.4$ using Milne's method.

c) Find the first and second derivatives of the function given below at the point $x = 1.2$:

x	1	2	3	4	5
y	0	1	5	6	8

Answer any two parts of the following.

a) In a partially destroyed laboratory record of an analysis of correlation

data, the following results only are eligible: $\sum x^2 = 9$,

regression equations: $8x - 10y + 66 = 0$, $40x - 18y = 214$. What were (i) the mean values of x and y (ii) the standard deviation of y (iii) coefficient of correlation between x and y .

b) Fit a second degree parabola to the following data :

x	1	2	3	4	5
y	1090	1220	1390	1625	1915

c) Calculate the first four moments of the following distribution about the mean hence find μ_1 and μ_2 :

x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

Sub Code: AHT- 006 ROLL NO..... ODD SEMESTER EXAMINATION,
2024 – 25

2nd Year (III Sem) B.Tech.

Advanced Applied Mathematics

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Find the Laplace transform of $f(t) = \cos t$ (5 marks) (ii) Find the Laplace transform of $f(t) = e^{-2t} \sin 3t$ (5 marks) b) Using Convolution theorem find the inverse Laplace transform of $F(s) = \frac{1}{(s^2+1)(s^2+9)}$ (10 marks) c) Using Laplace transform to solve the differential equation $y'' + y = 2e^{2t}$ with $y(0) = 2$ and $y'(0) = -1$ (10 marks)
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Find the Fourier transform of: $f(x) = \begin{cases} 1, & x < 1 \\ 0, & x > 1 \end{cases}$ (5 marks) (ii) Find the Fourier transform of: $f(x) = \begin{cases} 1 - x^2, & x \leq 1 \\ 0, & x > 1 \end{cases}$ (5 marks) b) Find the Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$. Hence derive Fourier sine transform of $f(x) = \frac{x}{1+x^2}$ (10 marks) c) Using finite Fourier transform, solve $y'' + y = 0$ given $y(0, t) = 0, y(4, t) = 0$

	and $\frac{\partial^2 z}{\partial x^2}(x, 0) = 2x$ where $0 < x < 4, t > 0$. (10 marks)
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Solve the equation $x^3 - x^2 - 1 = 0$ using the Newton-Raphson method. Perform three iterations starting with the initial guess $x_0 = 1.5$ (5 marks) (ii) Compute the value of $\frac{\partial^2 z}{\partial x^2}(x, t)$ for $x = 2.5$ from the following table: $\frac{\partial^2 z}{\partial x^2}$ 1 2 3 4 $\frac{\partial^2 z}{\partial x^2}(x, t)$ 1 8 27 64 Using Lagrange's interpolation formula. (5 marks)

	b) Find a real root of the equation $x^3 - 2x^2 - 5 = 0$ by the method of false position correct to three decimal places. (10 marks) c) Using Newton's forward difference interpolation formula, find the polynomial for the given data: (10 marks) $\frac{\partial^2 z}{\partial x^2}$ 0 1 2 3 $\frac{\partial^2 z}{\partial x^2}(x, t)$ 1 2 5 12
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Q 4.	Answer any two parts of the following. (10x2= 20) 6 a) (i) Evaluate $\int_0^2 \frac{1+x^2}{2} dx$ by using simpson's 1/3 rule. (5 marks) (ii) Evaluate $\int_0^2 \frac{1+x^2}{2} dx$ by using Trapezoidal rule. (5 marks) b) Use Runge-Kutta method to find y(1.2) in step size h=0.1 given that $\frac{dy}{dx} = x^2 + y^2$ with y(1)=1.5. (10 marks) c) Given $y'' = 1$ $y(0) = 2, y(0.2) = 2.0933, y(0.4) = 2.1755, y(0.6) = 2.2493$, find $y(0.8)$ by Milne's predictor corrector method. (10 marks)
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) By the method of least squares, find a straight line that best fits the following data points. x 0 1 2 3 4 y 1.0 2.9 4.8 6.7 8.6 (5 marks) (ii) The regression equations calculated from a given set of observations for two random variables are $\hat{y} = -0.4x + 6.4$ and $\hat{x} = -0.6y + 4.6$ Calculate r, r^2 (5 marks) b) Calculate coefficient of correlation from the following data: x 1 3 5 7 8 10 y 8 12 15 17 18 20 (10 marks) c) Fit a second degree parabola to the following data: x 1 3 4 6 8 9 11 14 y 1 2 4 4 5 7 8 9

	(10 marks)
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**Sub Code: AHT006/BAST401 ROLL NO..... EVEN SEMESTER
EXAMINATION, 2024 – 25**

Second Year (IV Sem) B.Tech.

ADVANCED APPLIED MATHEMATICS

MATHEMATICS III/IV

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed

and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) ∞ sin t ∫ – t e dt . a) (i) Evaluate t 0 (ii) State and prove Convolution theorem. 2 ss ++ 23 . b) Find the inverse Laplace transform of 22 ()() s s s s + + + + 2 2 2 5 c) Using Laplace transformation, solve the following differential equation: 2
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$$|f| \leq$$

$$x^x$$

$$f(x)$$

$$\left. \begin{matrix} () \\ = \end{matrix} \right\}$$

b) Find the Fourier transform of

$$|f| > \left(\text{Hence, evaluate} \right)$$

$$0, 1$$

$$x$$

$$\infty$$

$$\sin \cos$$

$$x^x x^x dx$$

$$\left. \begin{matrix} | \\ | \end{matrix} \right\} \int$$

$$\cos .$$

$$3$$

$$x$$

$$2$$

$$0$$

$$|f| <$$

$$1, 1$$

$$x$$

$$() .$$

$$\left. \begin{matrix} f(x) \\ = \end{matrix} \right\}$$

c) Find the Fourier integral representation of the function

$$|f| > \left(\text{Hence, evaluate} \right)$$

	$0, 1$ x ∞ $\sin \cos^{\sin}_{(ii)}.X$ $d\omega \omega \omega$ $\int \int$ $\omega \omega$ (i) $\omega \omega$ 00
Q 3.	Answer any two parts of the following. (10x2= 20) 3 cos 1 xx =+correct to four a) (i) Using Newton-Raphson method, find the real root of the equation decimal places.

	(ii) Construct the forward difference table, given that x 5 10 15 20 25 30 y 9962 9848 9659 9397 9063 8660 24 $_{10,5} \Delta \Delta y y$. and point out the values of b) From the table, estimate the number of students who obtained marks between 40 and 45. Marks: 30-40 40-50 50-60 60-70 70-80 No.of students 31 42 51 35 31 $f(8)$and $f(15)$from the c) By means of Newton's divided difference formula, find the values of following table:
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	$x^4 - 5x^7 + 10x^{11} - 13fx()$ $\frac{48}{100} - \frac{294}{900} + \frac{1210}{2028}$
Q 4.	Answer any two parts of the following. (10x2= 20) 6 3 1 $\int$ dx . a) (i) Using Simpson's 8rule on integration, evaluate 1 $+ x$ 0 6 dx $\int$ by using Simpson's one-third rule. (ii) Evaluate 2 1 $+ x$

	0 dy $y(0) = 1$ by Runge-Kutta method, from $= +$ with initial condition xy b) Solve the equation dx $x = 0$ to $x = 0.4$ with $h = 0.1$. , 2 $y \times y$ $= -$ in the range c) Apply Milne's method to find a solution of the differential equations $0 \leq x$ for the boundary condition $y = 0$ and $x = 0$.																																
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) The first four moments about mean of a frequency distribution are 0, 100, -7 and 35000. Discuss the kurtosis of the distribution. (ii) The first four moments of a distribution about the value 0 are -0.20, 1.76, -2.36 and 10.88. Find the moments about the mean and measure the kurtosis. b) The following table gives age (x) in years of cars and annual maintenance cost (y) in hundred rupees: <table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td></tr><tr><td>Y</td><td>15</td><td>18</td><td>21</td><td>23</td><td>22</td></tr></table> Estimate the maintenance cost for a 4 year old car after finding the regression equation. c) Fit a second degree parabola to the following data by least squares method: <table><tr><td>x</td><td>1929</td><td>1930</td><td>1931</td><td>1932</td><td>1933</td><td>1934</td><td>1935</td><td>1936</td><td>1937</td></tr><tr><td>y</td><td>352</td><td>356</td><td>357</td><td>358</td><td>360</td><td>361</td><td>361</td><td>360</td><td>359</td></tr></table>	x	1	3	5	7	9	Y	15	18	21	23	22	x	1929	1930	1931	1932	1933	1934	1935	1936	1937	y	352	356	357	358	360	361	361	360	359
x	1	3	5	7	9																												
Y	15	18	21	23	22																												
x	1929	1930	1931	1932	1933	1934	1935	1936	1937																								
y	352	356	357	358	360	361	361	360	359																								

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Odd Semester Examination, 2017-18

B.TECH. (SEMESTER-III)

COMPUTER BASED NUMERICAL &

STATISTICAL TECHNIQUES

Time: 02:00 Hours

Note:- Attempt **all** questions.

Max Marks: 50

1.

Attempt **any two** parts of the following:

(a)

(b)

(c)

[5x2=10]

Compute the Max. Relative Error for $R = 4xy^2z^3$, when absolute errors in x, y, z **be**

0.001 at $x = y = z = 1$.

1

Find a real root of $x^3 - x = 1$ between 1 and 2 by Bisection method. Compute three iterations.

Using Newton-Raphson method, find the square-root **of** 12 Correct upto 3 Decimal places.

Attempt **any two** parts of the following:

[5x2=10]

2.

(a)

Prove that

(i)

$$A \log f(x) = \log [1 + A f(x) / f(x)]$$

(ii) $E = 1 + A$

3.

(b)

Using Newton's Forward difference formula, to estimate $f(3.2)$, when

X: 0

1.

2

3

4

5

6

f(x): 2

4

10

16

20

24

38

=

=

(c) Use Stirling's formula to find y_{35} , given $y_{20} = 512$, $y_{30} = 439$, $y_{40} = 346$, $y_{50} = 243$.

Attempt **any two** parts of the following:

$$[5 \times 2 = 10]$$

1

(a)

Find

dx by Trapezoidal & Simpson's 01+ X

rule.

3

TCS-302/1620

(1)

[P.T.O.]

4.

(b)

The distance covered by an athlete for the 50 meter race is given in the following table:

Time (sec)

0

1

2

3

4

5

6

Distance(met.): 0 2.5 8.5 15.5 24.5 36.5

50

Determine the speed of athlete at 5sec. correct to two decimals.

Use Runge -kutta method to approximate y when $x = 0.2$, given that $y = 1$ when

(c)

$x = 0$ and

dy

dx

$=x+y$.

Attempt **any two** parts of the following:

[10x2=20]

(a)

Obtain the line of regression of y on x for the data given below:

X: 1.53 1.78

2.60

2.95

3.42

Y: 33.50 36.30 40

45.80

53.50

saen (b) n

Fit a straight line to the following data using Principle of Least square:

X: 1

2

3

y: 2.4

3

3.6

4

5

LO

4

6

8

6

10 (c)

5.2

Evaluate $\log x \, dx$, by Numerical Integration.

dy

Compute

at $x=1.1$ using following table:

dx

X: 1

2 3 4

5

y: 1

1 8 27

64

125

--X-----

TCS-302/1620

(2)

TCS 305

23

Printed Pages: **3**

Roll No. to be filled in your Answer Book

Roll No.

B. Tech III Semester Examination Dec 2014 Computer Based Numerical Technique

Time: 3.00 Hrs]

[Max. Marks: 75

no bat

Note: Attempt all questions, the marks assigned to each question is indicated at question itself.

Q1. Attempt any three

(5X3)

(a) Find the positive real root of $\cos x = xe$ using "Regula Falsi methods".

(b) Using Picard method, solve
up to third approximation.

$\frac{dy}{dx} = ex - y^2$ with $x=0$ and $y_0=1$

dx

(c) Find the positive root of $x^3 - 2x - 5 = 0$, correct to three decimal places using Newton's Raphson method

-

(d) Find the root of

1.

dx

2

by Waddle's rule.

$1+x-$

(e) show that the rate of convergence of newton Raphson method is 2.

Q2. Attempt any three

(5X3)

(a) Find the missing figures in following table:

X: 0 5

Y: 7 Y: 7 11

10 15 20

25

18

32

0

(b) Find the positive **root** of $3x=\cos x+1$, correct to three decimal **places using Iteration** method

(1)

TCS 305/1680 / 4

(c) Find the positive root of $3-2x-5=0$ correct to three decimal pe, places using Bisection Method.

(d) Fit the straight line to the following data:

(e)

X y

4

5

6

1 2 3

1200 900 600 200 110 50

The two regression equation of the variable x and y are $x = 19.13 - 0.87y$ and $y = 11.64 - .50x$

Find co relation coefficient and also regression coefficients.

Q3. Attempt any two

(7.5X2)

(a) What is statistical quality control? Define various control charts.

(b) Find $f(1.22)$ from the Gauss Backward method :

X F(x)

1.0

0.841

1.1

1.2

0.891 0.932 0.963 0.958

1.3 1.4

(c)) If $f(x) = 1/x$, then show that $f(a,b,c,d) = -1/(abcd)$

Q4. Attempt any two

(7.5X2)

(a). Find $f(1.22)$ from the Gauss forward method :

(b)

(c)

X

1.0

1.1

1.2

1.3

1.4

F(x) 0.841 0.891 0.932 0.963 0.958

Find **the cubic** polynomial using Bassel's rule which takes the following values

X

F(X)

3

168

7

9

10

120

72

63

From the following table:

X

1.7

1.74

1.78

1.82

1.86

sinx

0.9916 0.9857 **0.9781**

0.9691

0.9584

dy

(A)

-

Find **at** at $x=1.74\text{m}$

dx

(2)

(B)

Find

dx

24.2

$d^2 y$ at $x = 1.74$

(7.5X2)

Q5.Attempt any two

(a) Find the cubic polynomial using **Strling formula** which takes the following values

X

F(x)

3

168

7

9

120

22

10

72

63

(b) Using the Gauss Seidel method, solve the following equation:

$x+4y-z=-5$; $x+y-6z=-12$; $3x-y-z=4$

$4x+5$

1

(c) Find

$-dx$

(A)

(B)

By the Simpson's 1/3rd method.

By the Simpson's 3/8th method3

(3)

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B.TECH.(CS/IT)(SEMESTER-III)

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(e)

COMPUTER BASED NUMERICAL STATISTICAL

TECHNIQUES noitaald tart svor

Time: 03:00 Hours

Max Marks :50

1.

Attempt **any** four questions:

(a)

Derive Lagrange's Interpolation Formulae.

(b)

Find the real root of the equation $2x - \log x - 7 = 0$ using iteration method.

(c)

Find the rate of convergence for Regular-Falsi method.

(d)

(e)

[4x2.5=10]

Find the absolute, relative and percentage errors if x is rounded-off to three decimal digits. Given $x = 0.005998$.

Find a real root of $x^3 - x - 1$ between 1 and 2 by bisection method. Compute five iterations.

(f)

Evaluate

$$\int \frac{dx}{(1+x^2)}$$

using Simpson's 3/8 rule taking $h=1/6$.

2.6 89 Attempt **any four** questions: trigiera & no pinqui eo a to laskontoy enT

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$$[4 \times 2.5 = 10]$$

(a) Using the method of false position, find the root of equation $x - x - x - 1 = 0$ upto four decimal places.

(b)

Three approximate values of number $1/3$ are given as 0.30, 0.33 and 0.34. Which of these three is the best approximation?

TCS/TIT-302/2380

(1)

[P.T.O.]

(c)

Solve by iteration method: $\sin x = (x+1)/(x-1)$

(d)

What are the statistical quality control techniques? Discuss the objectives and advantages of statistical quality control.

of

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(e) Find the most plausible values of x and y from the following equations.

JADITBITAT? JADI

$$X+y=4, x-y=2, x+2y=4, x-2y=1$$

BTUAMOS

(f) Prove that bisection method is always converges.

3.

3.

Attempt any two questions:

[2x5=10]

(a) Find the root of the equation $x.e^x - \cos x - 0$ in the interval $(0,1)$, correct to the three decimal places using Regular-Falsi method.

(b) Apply Bessel's formula to find the value of $f(27.4)$ from the table:

Somex hoia 25

26

no 27

28

29

30

F(x)

4.000 3.846

3.704

3.571

3.448

3.333

sert (c) to-t Prove the Newton's Divided Difference Interpolation formula.

4.

Attempt any two questions.

avit storio, bor no

(a)

one i neowled

[2x5=10]

Use Gauss's forward formula to find a polynomial of degree four which takes the following values of the function $f(x)$.

X

F(x)

1

1

2

3

-1

1

-1

5

1

fotea (b) The velocities of a car running on a straight road at intervals of 2 minutes are given below:

Time (in minutes

0

2

4

6

8

10

12

HSTY- Velocity(in km/hr)

0

22

30

27

18

7

0

Apply Simpson's rule to find the distance covered by the car.

TCS/TIT-302/2380

(2)

5.

(c)

Find the missing figures in the following table:

X

x 2

2.1

2.2

2.3

2.4

2.5

2.6

y

0.135

0.111 0.100

0.082

0.074

COMPU

Attempt **any two** questions.

[2x5=10]

(a) Assuming that the following values of y belong to a polynomial of degree 4,
compute the next three values:

Time: 03:00

X

0

1

2

3

4

5

6

7

Y

1

-1

1

-1

1

(b)

(c)

Given the following table. Find $f(x)$ as a polynomial in powers of $(x-5)$ using Newton's divided difference formula.

X

0

2

3

4

7

9

F(x)

4

26

58

112

466

922

What do you understand by Stirling's formula? Discuss Stirling's formula along with its complete derivation.

TCS/TIT-302/2380

(3)

X

Sub Code: CST021 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2024 – 25

.207 | 18/06/2025 12:45:55

2nd Year (IV Sem) B.Tech.: CS&E/AI&ML/IT

COMPUTER NETWORKS

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	ST021_005 103.170.7⁰:45:5⁵ Answer any two parts of the following. (10x2= 20) a) (i) What are the design issues of network architecture? Explain. (ii) Discuss the devices of each layer of OSI reference model. b) What is OSI model? Use the network architecture knowledge to compare TCP/IP and OSI model. c) Define transmission media. Explain fibre optic cable in brief. Also explain the advantages of optical fibre over twisted pair and coaxial cable.
Q 2.	103.170.70.207 18/06/2025 1² Answer any two parts of the following. (10x2= 20) C a) (i) Discuss the Point-to-Point protocol with the help of frame format. (ii) What is piggybacking? Suppose we need a dataword of at least 7 bits. Calculate the values of k and n that satisfy this requirement. b) Explain different types of persistence method. How CSMA/CD protocol is different from CSMA/CA protocol? Explain. c) Explain Selective Repeat ARQ protocols. What are the advantages of Selective Repeat ARQ over Go Back-N ARQ protocol?
Q 3.	CST021_005 Answer any two parts of the following. (10x2= 20) a) (i) How is the BOOTP different from DHCP? (ii) Describe the problem of count to infinity associated with distance vector routing technique. 5 12:45:5⁵ b) What are different routing algorithms? Write the implementation of shortest path routing. c) Explain IPv6 header format in detail. Write advantages of Next-generation IPV6 over IPV4
Q 4.	3.170.70.207 18/06/20² Answer any two parts of the following. (10x2= 20) a) (i) What do you understand by Quality of Service? Explain various Quality of Service parameters. (ii) Enumerate how the transport layer ensure that the complete message arrives at the destination and in the proper order. b) Differentiate TCP and UDP in context of the header format.

	c) What is congestion? Briefly describe the techniques that prevent congestion.
Q 5.	cst021_005 10 Answer any two parts of the following. (10x2= 20) a) (i) Explain Asymmetric cryptography. Also write the steps used in RSA algorithm. (ii) What are firewalls? Why they are needed? Explain the different types of firewalls. b) Elaborate about TELNET. Discuss its working procedure in detail. c) Discuss the features of HTTP and also discuss how HTTP works.

WATER MARK (FOR SECURITY REASONS) CST021_005 | 103.170.70.207 | 18/06/2025 12:45:55

TCS-401 abortem Ingeffia melysil Sig) **2050**

Even Semester Examination 2017-18

B.TECH (SEMESTER-IV)

COMPUTER ORGANIZATION

Time: 03:00 Hours

Note: Attempt ALL question. All Question carry equal marks.

Max Marks: 100

1.

Attempt **any four** parts of following: legh

(5x4=20)

A.

What do you understand by stack frames? Discuss their use in sub-routines.

B. Which type of I/O devices are interfaced through DMA? Explain the Bus-arbitration process used for DMA?

C.

Briefly distinguish between textual data, pictorial data, audio& video data. And also explain the importance of ASCII code?

D.

Explain the procedure to multiply Two N bit numbers using hardware and booth algorithm?

E.

Simplify the following Boolean function F together with the don't care conditions'd' in sum-of-product forms and

$F(w,x,y,z)=(0,1,2,3,7,8,10)$ $d(w,x,y,z)=[(5,6,11,15)]$?

product-of-sums form

F.

What is decoder? Design 2 to 4-line decoder with NAND gates.

2.

Attempt **any four** parts of following:

(5x4=20)

isan A.

What is shift register? Explain the general capabilities of shift registers.

B.

What is Cache memory? Explain different types of mappings from main memory to cache memory.

TCS-401/2360

(1)

[P.T.O.]

C.

D.

E.

What is Asynchronous data transfer? Explain different methods used for asynchronous data transfer.?

What is fixed point representation of integers? Explain different ways to represent signed integers?

Describe SDRAM and DDR SD RAM operations for data transfer between main memory and cache memory systems.

F.

Differentiate between Micro operation and Macro Operation with an example.

3.

Attempt **any four** parts of following:

(5x4=20)

A. Differentiate Micro programmed control and Hardwired control.

B.

What is the impact of the cache on overall performance of the computer?.

C.

What do you understand by the term peripheral? Explain with some examples?

D.

"Instruction Set Architecture has impact on the processors micro architecture" support this statement with proper reasoning.

E.

F.

Explain different functional units of a digital computer with neat sketch?

Discuss the advantages, disadvantages, and applications of

(i)

Excess 3 code

(ii)

Gray Code (Illustrate with one example each).

o

4.

Attempt **any two** parts of following:

(10x2=20)

A.

Perform floating point addition using the numbers 0.5 and 0.4375 use the floating point addition algorithm.

B.

Explain the multiplication of positive numbers using array multiplier with a neat sketch.

C.

What is virtual memory? With the help of neat sketch explain the method of virtual to physical address translation.

TCS-401/2360

(2)

(10x2=20)

5.

Attempt **any two** parts of following:

A.

Explain the READ and WRITE operations in Associative Memory.

B.

C.

Draw the block diagram of a DMA controller and explain its functioning?

Discuss any five key differences between subroutine and interrupt service routines.

X

TCS-401/2360

(3)

TCS-401/TIT-401 des bo 1024/1025

Even Semester Examination 2018-19

B. TECH. (ECE/CSE/IT) (SEMESTER-IV)

Time: 03:00 Hours

COMPUTER ORGANIZATION

Max Marks :100

Note: A, B, C, D, and E are the five sections in the question paper. Attempt four parts from A,

four parts from B, four parts from C, two parts from D, and also two parts from the section E.

SECTION A

Attempt **any four** questions from the following. Each question carries **equal** marks.

[4x5=20]

1.

What is bus? Draw and describe the bus architecture for a digital computer.

2.

State the differences between serial and parallel transmission.

3.

What is a Von Neumann architecture? Also, discuss the Von Neumann bottleneck.

4.

Discuss your understanding of stack memory.

5.

Apply Booth's algorithm to multiply the two numbers (6) 10 and (-9) 10. Assume the multiplicand and multiplier to be 5-bits each.

SECTION B

Attempt **any four** questions from the following. Each question carries equal marks.

1.

Define the cache memory along with its significance.

2. Differentiate between write-through and write-back mechanism.

[4x5=20]

3.

Briefly explain the IEEE 754 standard format for floating point representation.

TCS-401_TIT-401/2380

(1)

MEESHONT [P.T.O.]

4.

State the differences between instruction-pipeline and arithmetic-pipeline. Ta

5.

What are the priority interrupts? Briefly explain any one interrupt priority scheme.

SECTION C

(VI)

Attempt **any four** questions from the following. Each question carries equal marks.

[4x5=20]

1. With a neat and clean diagram, explain the internal organization of a $2M \times *$ dynamic memory chip.

2.

Perform the following operation on the 5-bit signed numbers using 2's complement representation system. Also indicate if overflow has occurred.

(i)

$(-9) + (-7)$

(ii)

$(+7) - (-8)$

3.

Define exception and explain two of its kinds.

4.

Give the control sequence for the execution of instruction ADD (R3), R1.

5.

Define the following:

(i)

Memory latency

(ii)

Memory bandwidth

(iii)

Hit rate IS

(iv)

miss penalty

SECTION D

Attempt **any two** questions from the following. Each question carries equal marks.

1.

[2x10=20]

What is virtual memory? With a diagram, explain how virtual memory address is translated.

TCS-401_TIT-401/2380

(2)

2.

3.

Let you have a 256 MB main memory, and a 1 MB cache memory. The address space of the concerned processor is 256 MB. Block size is 128 bytes and there are 8 blocks in a cache set.

With the aforementioned specification, determine the size of sub-fields (in bits) in the address for direct mapping, associative and set associative mapping cache schemes.

Explain the parallel I/O interface with a block diagram.

SECTION E

Write short notes on **any two** of the following. Each carries equal marks.

1.

Addressing modes

2.

RISC

3.

Cache Coherence

TCS-401 TIT-401/2380

(3)

X

[2x10=20]

CST007

**Sub Code: CST 007 ROLL NO..... EVEN SEMESTER EXAMINATION,
2024 – 25**

II Year (IV Sem) B.Tech.

Computer Organization & Architecture

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Describe the register organization in CPU. (ii) Differentiate between RISC and CISC architecture. b) Define data manipulation instructions. Consider a processor with 16 general purpose registers uses a 32- bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, describe the maximum number of unique opcodes possible for every addressing mode. c) Define stack CPU organization. Consider the following statement executed on a stack CPU 'X= (A+B) / (C*D)' variables in the memory. i. How many machine instruction required. ii. If a processor contain 8 bit op-code and supports 64 KB memory. How much space is required for program?
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Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Define control unit in CPU. (ii) Describe address lines, control lines and data lines in bus organization. b) Explain signed operand multiplication algorithm. Consider a 4-bit data as (-7) * (-3) as multiplicand and multiplier. Describe recoded multiplier and final product using booth's multiplication algorithm. c) Describe the advantages of micro-programmed approach over hardwired approach in Control unit. Consider a hypothetical CU supports five groups of mutually exclusive control signals as given {G1, G2, G3, G4, G5}; control signal {1, 10, 20, 30, 40}. Describe how many bits saved using vertical over horizontal programming.
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Define USB.

CST007

	(ii) Explain Interrupt Driven I/O transfers. b) Explain the working of DMA along with detailed diagram. c) Define Interrupts. Explain its role in process state transitions.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) Differentiate between shared and distributed memory multiprocessors. (ii) Define synchronization. b) Explain the concept of pipelining. Consider a 5-stage instruction pipeline has stage delays of 180,250,150,170, and 250 respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. Describe the time taken to process 1000 instructions in microseconds. c) Define hazards. Explain the different types of hazards in detail.

Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Explain hierarchical memory structure in terms of speed, size, cost. (ii) Consider 4B cache memory (initially empty) with following memory block references 4,5,7,12,4,5,13,4,5,7. Describe HIT ratio using FIFO and LRU page replacement policies. b) Define locality of reference. Consider a 16-way set associative cache of 64KB organized into a 64B blocks. Processor generates 40-bit address to access the data. In cache controller each tag is comprising of 1 valid bit, 1 modified bit and 2 replacement bits. Describe tag memory size. c) Explain the problem of cache coherence and how it can be removed. Also give a suitable example.
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Sub Code: BCST404/BEET404/BITT404 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2023 – 24

**IIInd yr B.Tech. – Computer Science & Engineering/Information Technology
Computer Organization and Architecture**

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) What is RTL? Draw logic diagram for “D3.T4: R1 R2 b) Explain steps to fetch a word from memory with suitable example and diagram. c) What is stack organization? Explain with suitable example. d) What is memory hierarchy? Why it’s needed? e) What is pipelining? Explain with example. f) What is instruction cycle? Explain with suitable example.	5x4=20
Q 2.	Answer any four parts of the following. a) Design 4 bit Binary incrementer. b) What is auxiliary memory? Explain with suitable example. c) What is multiprocessor? d) What is program control? e) What is prefetching in architecture? f) What is interrupt? How architecture handles it?	5x4=20
Q 3.	Answer any two parts of the following. a) Design 4 bit arithmetic circuit. b) What is Direct Memory Access? Discuss its functionality with suitable diagram c) What is cache memory? Discuss all mapping methodology with suitable diagram and example	10x2= 20
Q 4.	Answer any two parts of the following. a) What is addressing mode? Discuss displacement addressing mode with suitable example and types. b) Design four 4-bit registers shared common bus using multiplexers. c) Explain Booth’s multiplication algorithm with suitable example and flowchart.	10x2= 20

Q 5.	Answer any two parts of the following. a). What is cache coherence? Suggest a solution to solve it? b) What is hardwired control? Explain with suitable example and diagram. c) Design 4-bit carry look ahead adder.	10x2= 20
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Sub Code: CST-007 ROLL NO.....

EVEN SEMESTER EXAMINATION, 2023 – 24

2nd yr B.Tech. – AI&ML/CS&E

Computer Organization and Architecture

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) State about parallel processing. Explain Flynn's classification. b) Differentiate between hardwired and microprogrammed control unit. c) An 8-bit register contains the binary value 10011100. What is the register value after an arithmetic shift right? Starting from the initial number 10011100, determine the register value after arithmetic shift left and state whether there is an overflow. d) State about pipeline hazards. Classify pipeline hazards. e) What do you mean asynchronous data transfer? Define strobe control method in brief. f) Explain CISC and RISC with example.	5x4=20
Q 2.	Answer any four parts of the following. a) Calculate the page fault for a given string with the help of LRU page replacement algorithm, size of frames = 4 and string 1 2 3 4 2 1 5 6 2 1 2 3 7 6 2 3 1 2 3 6. b) Point out the issue occurs in distributed shared memory. c) Explain floating point representation with suitable example. d) Define pipeline. Outline the advantages and disadvantage of it. e) Outline the need of I/O addressing. Compare I/O mapped I/O with Memory mapped I/O. f) Explain different mode of data transfer.	5x4=20
Q 3.	Answer any two parts of the following. a) Write short notes on following i. Locality of reference. ii. Interrupts. b) An address space is specified by 24 bits and the corresponding memory space by 16 bits. i. How many words are there in the address space? ii. How many words are there in the memory space?	10x2=20

	iii. If a page consists of 2K words, how many pages and blocks are there in the system? c) Explain in detail multiple bus organization with the help of diagram.	
Q 4.	Answer any two parts of the following. a) Why does DMA have priority over the CPU when both request a memory transfer? Discuss burst transfer, cycle stealing and DMA transfer in CPU. b) Define micro program sequencer. With block diagram discuss the working of micro program sequencer. c) Why a number of addressing mode is needed? By taking a suitable examples explain the following addressing modes: i. Direct	10x2= 20

	ii. Index iii. Relative iv. Immediate v. Register.	
Q 5.	Answer any two parts of the following. a) Show the multiplication process of (20) X (-19) using Booth's algorithm. b) Define cache memory. Explain associative and set associative mapping with suitable example. c) Draw and explain the flowchart for instruction cycle with neat diagram.	10x2= 20

5

Write Short notes on any four of the following: (a)

5×4=20

Differentiate between the working of static and dynamic memory. Also discuss the difference between SRAM and DRAM.

(b) Design a synchronous modulo-12 counter using NAND gates and T flip-flops.

(c) Design a hazard-free, D-type flip-flop using asynchronous circuit design techniques. It may be assumed that the output will take on the value of the input on the trailing edge of a clock pulse.

PAPER ID: 3074

TEC-302

Printed Pages: 4

Paper ID and Roll No. to be filled in your Answer Book Roll No.

B. Tech.

(SEM. III) (ODD SEM.) (REG. & BACK) EXAMINATION, 2012-13 **DIGITAL ELECTRONICS & DESIGN ASPECTS** [Total Marks: 100

Time: **3 Hours**]

Note: (1)

Attempt all questions

(2)

Marks are indicated at the end of each section

1

Attempt any four parts of the following:

5×4=20

(a)

Determine the base b in each of the following

cases:

(i)

$$(361)_{10} = (551)_b$$

(ii)

$$(859)_{10} = (5B7)_b$$

(b)

Write the 8-bit signed magnitude, 2's complement

(c)

and 1's complement form of the following decimal numbers:

+ 119

(ii) -77

Write out the first ten numbers for decimal weighted codes: 7, 4, 2, 1

(d) Using the theorems of Boolean algebra simplify the following expression:

(e)

$F(A, B, C, D) = B + BCD + BCD + AB + AB + BC$ What is the difference between a Latch and a

flip-flop?

(f)

What is race around condition and how it is overcome?

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1

[Contd...

2

Attempt any Two parts of the following:

(a)

5×4=20

Construct a truth table for the following functions and from the truth table obtain an expression for the inverse functions:

(i)

$$f(A,B,C)=A+BC$$

3

3074]

(ii)

$$f(A, B, C)=(A+B)(A+B+ +C)$$

(b)

For the following two 4-variable functions

$$f_1 = A+B+C+D$$

$$f_2 = A+C+BD$$

how many of the input minterms are included in each of these functions and how many are not? What are the minterm expressions for the two functions? Simplify both functions using the theorems of Boolean algebra.

(c) Simplify the following three-variable Boolean functions algebraically and express them in terms of product of maxterms:

(i) $f_1 = \Sigma(1,2,5, 6)$

(ii) $f_2 = (0, 1, 2, 3, 7).$

Attempt any Two parts of the following: (a)

10×2=20

Minimize the following function and implement the minimized function using only NAND gates: $f(A, B, C, D) = (0, 2, 8, 9, 10, 12, 13, 14)$

2

(b) Analyze the circuits shown in fig. 1 to produce Boolean algebraic expressions for the circuit outputs.

A

B

I

C

D

D

E

F

D

G

D

Fig. 1

(c) Implement the following function using only NOR gates having a maximum fan-in of three:

$$f = (A+B)(C+D)(B+C)(A+D)(A+C)$$

4

Attempt any two parts of the following:

[**Contd...**

3074]

10×2=20

(a) Implement the following 3-variable Boolean functions using 4-input multiplexers :

$$f = \Sigma(0, 2, 3, 5, 7), \text{ control variables A and B.}$$

(b) Design a circuit for converting from the 8421 code to the 5421 code and implement the design with 4-to-1 multiplexers.

(c) Develop a 3-to-8 line decoder using NOR gates only, and draw its logic diagram.

3

[**Contd...**

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PAPER ID:1051

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Paper **ID and Roli No. to be filled in your Answer Book**

Roll **No.**

Time: **3 Hours]**

B. **Tech.** VOM

(SEM. IV) EXAMINATION, 2012

MICROPROCESSORS

VOM

[Total Marks: **100**

99signal vidrosees 3808 m brs minogls ns sthW (d) **Note** :by Attempt all Questions. All questions carry equal marks.

(a)

(b)

Jad W

1 STo Attempt any Four parts $5 \times 4 = 20$

of the following: nem Explain the evaluation of microprocessor in brief. Draw the functional block diagram of Intel-8085. (c) State the function of temporary registers of the 8085 ad sq 05-01 microprocessor.llo to an owns tom

2

22(d) State the features of 8086. Also explain the difference bas Obetween 8086 and 8085.

inco pri odisest

Define the term interrupt, exception, fault, and trap. List the four major processing units in an 80286 microprocessor and describe the function of each. onsdw Viiteul stubosong Sonam nesvied sisimeniftia Attempt any Two parts of the following om 10×2=20 (a) Draw typical 8086 minimum mode configuration and explain function of two signals used in minimum mode. (b) Describe memory mapped I/O with suitable example. sovho State its advantage over I/O mapped I/O.

unction of two signals

(c) Draw internal architecture of 8086 microprocessors. Also explain the function EU.

[OSSE]

[1201-20

DS-1051]

15

[Contd...

3

4

5

Attempt any 'Two parts of the following:

(a)

10×2=20 State any four addressing modes used in 8086 microprocessor. Identify Addressing modes used in each of the following 8086 instructions.

(i)

MOV BX,0354h

(ii) OS ADD AL,[BX+04J (VI ME2)

MOV AX,[BX+SI]IM

(iv) MOV AX,[BX+SI+04]

Coraham Istot

os

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(b) Write an algorithm and an 8086 assembly language program to arrange numbers in ascending order. What is memory segmentation? How memory segmentation is achieved in the 8086 microprocessor? and state

(c)

State advantages of memory segmentation..

Attempt any Two parts of the following:

10×2=20

(a) Explain the different modes of operation of 8255. Describe the control word format of 8255 in 1/0 and

bits 85R mode.19 mi osad ni

(b) Explain the various functional block diagrams of 8251.

(c)

bus ne

o

Differentiate between macro & procedure, justify where macro is suitable than procedure with example.

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Attempt any Two parts of the following: **taskyp**

10×2=20

sigma x Sidama the bagg viomem adinoes() (a) Describe the operation of a flash-type A/D converter. Poson are 2109007 What are its main advantages and disadvantages?

(b) Describe the operation performed by each of the following 8087 instructions.

(c)

(1) FLD TAX RATE

(11)

FMUL INFLATION FACTOR

(iii) FSQRT

(iv) FLDPI

FSTSW CHECK -ANSWER

(v)

(vi) FPTAN

Differentiate the processors 8087 and 8089. Explain how 8087 is interfaced to 8086.

DS-1051]

21

[Contd...

DS-1051]

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TCS /TIT- 401

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B. Tech. (Information Technology) (4TH Semester)

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Examination, 2015 obtain Examination, 2015 with **Computer Organization**/bish s

Time: 3.00 Hrs]

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[Max. Marks: 100

Note: All questions are compulsory.

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Q1. Attempt any four parts:

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(5×4 = 20)

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pniwollo) erit no seton horleenW

(a) What is shift micro operation? Explain logical shift, circular shift and arithmetic shift micro operation.

340 with

(b) Define bus arbitration. Also dis

Also discuss serial bus

arbitration with proper diagram.

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21 contains the

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(c) What do you mean by inter-register transfer?

Discuss Bus transfer.

(d) Explain addition and subtraction algorithm with the help of flow chart.

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(1)

(e) Multiply decimal -5 by decimal -6 using Booth's algorithm.

direct

(1) TCS/TIT-401/2800/8

0: espЯ being

(f)

Discuss IEEE standard for floating point number. Xoca lewan juo ni belli ed of of lo

Q2. Attempt any four parts: (5x4 = 20)

Ajo

helasms2 A) (veolonyos ngisimmoin!) noe! @ (a) What do you mean by Control unit? Describe briefly

a Hardwired control unit. uqmo

(b) Distinguish between a control memory and main

Text 00.8 :emiT memory used in a digital computer.

(c) What do you mean by the term Micro program

Sequence? Explain the role played by it.

(OS=Ax2)

(d) Write short notes on the following:

alox

itsnW. (5)

bris fine isluonio

(i)

Microinstruction

aud ishe Micro Program ichs and enite (d)

(ii)

msipsib neqorq ritw.noitentichs

(e) Explain wide-branch addressing and emultion. fretensil staigen 197 yd nsemi woy **ob** tortW

(1) Explain:

ari niliw m(i)hopFetching a word from memory

(ii) Storing a word in memory

herlo woll to gleri

(9)

a'diooE prizu a-lsmineb vd - lsmiseb vigiliuM

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(2)

mrithapis

Q3. Attempt any two parts:

AMD Jew ($10 \times 2 = 20$)

te the program to

(a) Write the

Me Rache

program to evaluate the arithmetic

sgrepiting Statement $X = (A + B * C) / (D - E * F + G * H)$:

tion of virt

euchsy 2(i).eit Using a general-register-type computer with three-address instructions.

ddress

(054 (ii) Using a general-register-type computer with

glen ert rit

(iii)

ton or noase

two-address instructions.

Using an accumulator-type computer with one-

otherearter-transfer

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overleaf

address instructions.

instructions. bold s to specification given in the table

lis enhot (iv) Using a stack-organ

Using a stack-organized computer with zero-

condi address operation instructions.

for

arx (b) An instruction is stored at location 300 with its

2920 erbe

301. The address field has

address field at location 301. The addres

the value 400. A processor register R1 contains the abow tool to exia bold s riw prices
Josib

number 200. Evaluate the effective address if the

bold,xeb addressing mode of the instruction is:

SWOR

Siem (i) 22 Directrolatis ablow bris

(ii) Immediate

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gribes to brow rose neverthens alid ynem woH

(iii) Relative (\$88)

Sanoitonul dini bobivib verit ens word bre

(iv) Index with R1 as index register

33

(v) Register indirect

(3)

(0)

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(c) (i)

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What is DMA scheme of data transfer?

Discuss its operating principle.

(ii) What is an interrupt? Describe its significance

ritiw letuamoo

for digital computers. Also discuss various

types of interrupts in brief.

Q4. Attempt any two parts:

(10×2=20)

(a) Explain the need of memory hierarchy with the help of a block diagram? What is the reason for not having one large memory unit for storing all

information at one place?

(0)

(b) A digital computer has a memory unit of 64K x 16

and a cache memory of 1 K words. The cache uses

A

direct mapping with a block size of four words.

(i)

How many bits are there in the tag, index, block,

and words fields **of** the address format?

(1)

(ii) How many bits are there in each word of cache,

and how are they divided into functions?

(4)

(1)

eroes sil

sit \$t1, \$t5, \$s2

\$s2word

bne \$t1, \$0, loop

Isuhiv to

(i)

t Draw a space-time graph showing the

hot smerte progression of the instructions for one iteration

of the loop through this pipeline.

Lazenbbs leoiayor th

epste evil

(ii) How many cycles elapse between the initiation

of successive iterations?

A

(b) Explain the concept of pipelining, instruction 19tansul-19teiperent or pribroos

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sesi enj ni nevi nousoffio

tashavo pipelining, RISC pipelining and arithmetic bisss pipelining.

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(i)

srit stubs Inter-processor Arbitration

(ii)

unit. blowieqi & noua wodnebienoo

Cache Coherence gool gniwollof

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block size of four

wgool

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B. Tech. (Information Technology) (4TH Semester)

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computer

Examination, 2015 obtenW Examination, 2015 with **Computer Organization**/bish s

Time: 3.00 Hrs]

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[Max. Marks: 100

Note: All questions are compulsory.

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Q1. Attempt any four parts:

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(5×4 = 20)

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pniwollo) erit no seton horleenW

(a) What is shift micro operation? Explain logical shift, circular shift and arithmetic shift micro operation.

340 with

(b) Define bus arbitration. Also dis

Also discuss serial bus

arbitration with proper diagram.

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21 contains the

grey the

nisloxa

(c) What do you mean by inter-register transfer?

Discuss Bus transfer.

(d) Explain addition and subtraction algorithm with the help of flow chart.

viomem ni blows gnino?

(1)

(e) Multiply decimal -5 by decimal -6 using Booth's algorithm.

direct

(1) TCS/TIT-401/2800/8

0: espЯ being

(f)

Discuss IEEE standard for floating point number. Xoca lewan juo ni belli ed of of lo

Q2. Attempt any four parts: (5x4 = 20)

Ajo

helasms2 A) (veolonyos ngisimmoin!) noe! @ (a) What do you mean by Control unit? Describe briefly

a Hardwired control unit. uqmo

(b) Distinguish between a control memory and main

Text 00.8 :emiT memory used in a digital computer.

(c) What do you mean by the term Micro program

Sequence? Explain the role played by it.

(OS=A_x²)

(d) Write short notes on the following:

alox

itsnW. (5)

bris fine isluonio

(i)

Microinstruction

aud ishe Micro Program ichs and enite (d)

(ii)

msipsib neqorq ritw.noitentichs

(e) Explain wide-branch addressing and emultion. fretensil staigen 197 yd nsemi woy **ob**
tortW

(1) Explain:

ari niliw m(i)hopFetching a word from memory

(ii) Storing a word in memory

herlo woll to gleri

(9)

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Q3. Attempt any two parts:

AMD Jew (10×2 = 20)

te the program to

(a) Write the

Me Rache

program to evaluate the arithmetic

sgrepiting Statement $X = (A + B * C) / (D - E * F + G * H)$:

tion of virt

euchsy 2(i).eit Using a general-register-type computer with three-address instructions.

ddress

(054 (ii) Using a general-register-type computer with

glen ert rit

(iii)

ton or noase

two-address instructions.

Using an accumulator-type computer with one-

otherearter-transfer

pied ov

overleaf

address instructions.

instructions.bold s to specification given in the table

lis enhot (iv) Using a stack-organ

Using a stack-organized computer with zero-

condi address operation instructions.

for

arx (b) An instruction is stored at location 300 with its

2920 erbe

301. The address field has

address field at location 301. The addres

the value 400. A processor register R1 contains the abow tool to exia bold s riw prices
Josib

number 200. Evaluate the effective address if the

bold,xeb addressing mode of the instruction is:

SWOR

Siem (i) 22 Directrolatis ablow bris

(ii) Immediate

512

gribes to brow rose neverthens alid ynem woH

(iii) Relative (\$88)

Sanoitonul dini bobivib verit ens word bre

(iv) Index with R1 as index register

33

(v) Register indirect

(3)

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02.

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(c) (i)

sitemdths srit

What is DMA scheme of data transfer?

Discuss its operating principle.

(ii) What is an interrupt? Describe its significance

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for digital computers. Also discuss various

types of interrupts in brief.

Q4. Attempt any two parts:

(10×2=20)

(a) Explain the need of memory hierarchy with the help of a block diagram? What is the reason for not 0195 r having one large memory unit for storing all

information at one place? sibbe

(0)

ati (b) A digital computer has a memory unit of 64K x 16

and a cache memory of 1 K words. The cache uses

etape 102250019 A 00 eulsy erit direct mapping with a block size of four words.

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erit ti szebbe svijets et sind

(i)

How many bits are there in the tag, index, block,

and words fields **of** the address format?

sisibammi (1)

(ii) How many bits are there in each word of cache,

evilsle?: (i)

and how are they divided into functions? 1912lgexebni es diwxobal (vi)

joeibnitzipeЯ

(4)

(1)

eroes sil

sit \$t1, \$t5, \$s2

\$s2word

bne \$t1, \$0, loop

Isuhiv to

(i)

t Draw a space-time graph showing the

hot smerte progression of the instructions for one iteration

of the loop through this pipeline.

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epste evil

(ii) How many cycles elapse between the initiation

of successive iterations?

A

(b) Explain the concept of pipelining, instruction 19tansul-19teiperent or pribroos pritsiago enilgiq

sesi enj ni nevi nousoffio

tashavo pipelining, RISC pipelining and arithmetic bisss pipelining.

s

visa (c)perWrite short notes on:a-brisanoitibroo

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(i)

srit stubs Inter-processor Arbitration

(ii)

unit. blowieqi & noua wodnebienoo

Cache Coherence gool gniwollof

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block size of four

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and words12 820 828 duen

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+882 828 ibbs

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TCS-404

172-A

Printed Pages: 4

Roll No. to be filled in your Answer Book

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B. Tech. Computer Science Engineering (4TH Semester) Examination, 2015 **Database Management System**

Time: 3.00 Hrs]

00 Hrs] by

Note: All questions are compulsory.

[Max. Marks: 100

1.

Attempt any four parts-

5*4 = 20

a)

Explain advantages and disadvantages of database system over conventional file system.

b)

Draw an ER diagram for a small marketing company database, assuming your own data requirements.

c)

Explain the distinctions among the term primary key, candidate key and super key with example?

8/d) What are the mapping cardinality constraints and participating constraints?

e)

Explain the database management system

architecture with neat diagram.

f)

Who is a DBA? What are the responsibilities of a

DBA?

(1)

TCS-404/1880/4

2.

Attempt any four parts-

a)

5*4 = 20

Write the tuple relational calculus expression to find

the number of employees working in **sales**

department in the given relation employee

Employee (SSN-no, NAME, DEPARTMENT)

001

b)

Define a NULL value? How do you retrieve null

values from the database?

c)

Consider the following relation

metay Employee (Employee-Name, Company-

Name, Salary)

I. Find the Total salary of each company

II. Find the employee name that is getting lowest

salary.

bne d) Given R(A, B, C, D, E) with the set of FDs F {AB->

mele

CD, ABC -> E, C->A}

I. Find any two candidate key of R

II. What is the normal form of R? Justify

80

(2)

e) Consider the following schema:

SUPPLIER (SUPPLIER_ID, SUPPLIER_NAME,

po SUPPLIER_ADDRESS)

PARTS (PART_ID, PART_NAME, COLOR)

CATALOG (SUPPLIER_ID, PART_ID, COST)

Write the following queries in relational algebra and SQL:

f)

I. Find the name of supplier who supply yellow parts.

II. Find the name of suppliers who supply both blue
and green parts.

Write short notes on:

I.

ul. DDL

II. DML

3.

Attempt any TWO parts-

a)

b)

Desc)

$10 \times 2 = 20$

What is normalization? Explain 2NF with insertion,
deletion and updation anomalies with an appropriate
bom benstab en terw example?

When is a functional dependency said to be trivial?

What do you mean by Natural join? Differentiate between Cartesian product and natural join with example.

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(3)

TCS-404/1880/4

4.

Attempt any TWO parts-

$10 \times 2 = 20$

5.

a) What is Transaction? List and discuss ACID properties of a Transaction. Define Transaction Log and what its function is.

b) What is a view? How can it be created? Explain
108

with an example.

c) When is a transaction said to be deadlocked?
ould no Explain the deadlock prevention methods with an example.

Attempt any two parts- Roast horle s $10 \times 2 = 20$

a) Explain the various ways in which concurrency control can be implemented in a database.

b) Discuss in detail about transaction concept and two nothsen phase commit protocol.

mon ails

c) What are deferred modification and immediate

Sisivite modification techniques for recovery? How recovery elitnens does takes place in case of failure in these

iw not techniques.

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TCS-404/1880/4

TCS 405

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Printed Pages:

Paper Code & Roll No. to be filled in your Answer Book

Roll No.

B.TECH

SEM IV (EVEN SEM) EXAMINATION 2014-15

MICROPROCESSORS

Time: 3 Hrs.

Max. Marks :100

Note: Attempt All Questions:

1.

Attempt any four questions

(5*4=20)

a)

Draw & discuss the internal block diagram Of 8086.

b)

c)

How does an 8086 differentiate between an **op** code and instruction data?

Draw the register organization of 8086 and explain application of each register?

d)

Explain the microprocessor evolution & types

e)

f)

Write a short note on (i) Pentium Processor (ii) Microcontroller Differentiate between the following microprocessor:

(i) 80286 (ii) 80386 (iii) 80486

2.

Answer any four questions

(5*4=20)

a)

Explain the Maximum Mode with Read Cycle Timing Diagrams.

b)

Write an assembly language program to find out the largest number from a given unordered array.

c)

Describe the different addressing modes in 8086.

d)

What is the difference between a near & far procedure?

e)

Explain the memory segmentation **in** 8086?

(1)

TCS-405 // 1720

f)

Differentiate between Immediate, Direct, and Register Addressing modes.

2

Attempt any two question

(10*2=20)

a)

Explain the General Bus Operation with Block Diagram.

b)

Write a program to add a data byte located at an offset 0500H in 2000H segment to another data byte at 0600H in the 3000H segment. Store the result at 0700H in the 0700H segment.

c)

Explain the 16 bit flag register in 8086.

Q4. Attempt any two Questions.

a)

b)

(10*2=20)

Explain the Interfacing 8086 with internal architecture of 8255 PIO and Interfacing 8086 with internal block diagram of 8279.

Explain the Interfacing 8086 with internal block diagram of 8253 and Interfacing 8086 with internal architecture of 8251 USART.

c)

How assemblers MACRO differ from procedure? Interface two 4k*8 EPROMS and two 4k*8 RAM Chips with 8086.

Q5. Answer any two questions. (10*2=20)

a)

Explain the interfacing of A/D Converter & D/A Converters?

b)

Explain the architecture of Numeric Processor 8087?

c)

Explain the architecture of I/O Processor 8089?

(2)

TCS-405 // 1720

TCS-403

Con **2232** NFA given in follow Printed Pages: 6

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Roll No. to be filled in your Answer Book viseMeri

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B. Tech. (Computer Science Engineering) uqlub (4TH Semester) Examination, 2015

op **Theory of Automata**

Time: 3.00 Hrs]

Note: All questions are compulsory

Q.1 (attempt any four)

P

[Max. Marks: 100

$5 \times 4 = 20$

Jolyne fometta) SD

a) Compare and contrast Finite Automaton, Push

priufto mspo

Down Automaton and Turing Machine?

b) Explain Context free grammar with suitable

jaom m

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eldslive pricexamples. sip priiwoolk

03

c)

d)

Differentiate between DFA and NFA using two suitable examples.

Prove or disprove regularity of language $L = \{0^n 1^n \mid n \geq 0\}$ using pumping lemma.

tsnt rice) What do you understand by ambiguous grammar?

ve neve Explain with suitable example.

□

belo

(1)

TCS-403/1820/6

eros not states Output

a esp belConstruct a Moore machine which is equivalent to

f)

the Mealy machine defined by table given below:

Present states For Input a=0

(prinsenigna

For Input a=1

state

output

q1

93

10

92

1

0

q2

94

1

92

0-

q, laH01comi

Aл

94

0

93

1

Q.2 (Attempt any four)-

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a)

$5*4=20$

What do you mean by halting problem of turing machine.

eldestis die

b) Explain Left most derivation and Right most derivation for following grammar by using suitable string.

owl prie ATM ons A70 neewed olsitost

$S \rightarrow S+G/G$

-zelqmaxe eldstiva

$G^* \rightarrow G^*Z/Z/id$

$Z \rightarrow (S^*G) / \text{Goniqmuq prizu}$ ($0 \leq$

Stemmc) pa Design DFA for language Lover $\{0,1\}$ such that every string in L start with 101 and end with **11**.

810587180-20T-

(2)

d)

Convert the NFA given in following table to its corresponding DFA. Here q0 is initial and q2,q4 are final states.

State

Input

b

a

→9%

9092

9391

91

9293

919%

X

92

939 2

94

93

909192

9% 9394

*

94

9%

9492 jams) \$0

rttiwester

Construct NFA for the set of all

Construct NFA for the set of all strings over the alphabet $\{0,1\}$, where 5th symbol from right is a '1'.

vd f)mm Define Deterministic finite automata and its tuples

dienst with suitable example.

Q.3 (attempt any TWO)-

hat is deck

a)

b)

of tale

bonite

916 P

c)

ue owf pnieu

$10 \times 2 = 20$

decidatio

Construct PDA that recognizes the language

$L = \{wcw \mid w \in \{a, b\}^*\}$

Construct Turing Machine which works as a copying machine for us esta muminim

novianst

aldet contenenent scor

smblus navig

$WE(a, b)^*$. pe19H woled nevig elds! yd

pengh

Find the regular expression corresponding to the finite automaton given below using Arden's theorem.

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TCS-403/000/6

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the Mealy r

iste 1,0

Present states

0

91

1612

90

0

1

Pop

1

q2

1,0 P

PP.P

Q.4 (attempt any TWO)-

02

a)

$10 \times 2 = 20$

What do you mean by Turing Machine? Explain with suitable example. per (10) jedsricis

b) 2 Design two parse trees for following grammar by

using two suitable strings having different length.

derivation for foll

using

OS SO $E \rightarrow [E+T]/T$

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$T \rightarrow T * F / F$

(S

$F \rightarrow \{E+F\}^* Flid$

ida) $www\} = J$

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(d

c)

Construct a minimum state automaton equivalent to
given automaton M whose transition table is defined
by table given below. Here q_0 is initial and
 q_1, q_2 are
final states.

in 1-start with

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misioonita nobiApniau waled nevie notsmotus etinit

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(4)

State

Input

b

a

9%

9%

93

91

92

95

92

93

94

93

9%

95

94

9%

96

97

95

91

96

91

93

9%

95

97

a)

Q.5 (attempt any TWO)-

$0*2=20$

What is decidability? Explain any two undecidable problems.

b)

Reduce following grammar and convert it to Chomsky Normal Form(CNF).

c)

$E \rightarrow E+T/T$

$T \rightarrow T* F/F$

$F \rightarrow id / (E)$

Explain pumping lemma for context free language with suitable example.

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BEET-405/BITT-405

Roll No.

Time: 3 Hours

Even Semester Examination, 2021-22 **Course Name : B.TECH**

Branch: CSE/IT

Semester: IV

Subject: Theory of Automata & Formal Languages

Max Marks: 100

Number of Printed pages: 2

Note: - Attempt all questions: All Questions carry equal marks

Q1. Attempt any four parts of the following

(5 x 4 = 20)

(a) What is a Finite Automata? Design a FA that accepts strings containing exactly four 1's in

every string over alphabet {0,1}.

(b) Give the complete hierarchy of Grammars with their recognizers as well as the form of

Production rules.

(c) Differentiate between Kleene Closure and Positive Closure.

(d) Give the Mathematical definition of NFA and state main difference between NFA & DFA. (e) Design a DFA for language L over {0, 1} such that every string in L start with 101 and end

with 11.

Q 2. Attempt any four parts of the following

(5 x 4 = 20)

(a) Design a DFA for the language "set of the string starting with 1 and number of 0's is divisible by 3". (b) Construct a NFA for set of all strings over alphabets {0,1} where 5th symbol from right is '1'. (c) Convert following Mealy Machine to an equivalent Moore machine. (Consider Q0 as the initial state)

Present **State**

I/P = 0

New State

O/P

New State

I/P = 1

O/P

□ Q0

Q0

1

Q1

0

Q1

Q2

Q3

222

Q3

1

03

1

1

0

88

Q2

1

1

Q2

(d) Remove useless symbols to simplify the given grammar:

$S \rightarrow aaA|B|C$

A $\rightarrow$ aB | s

B $\rightarrow$ Aa

C $\rightarrow$ cCD

D $\rightarrow$ dd

(e) Show that following grammar is ambiguous:

S $\rightarrow$ AB | aaB

A $\rightarrow$ a | Aa

B $\rightarrow$ b

P.T.O.

Q3. Attempt any two parts of the following

(10 x 2 = 20)

(a) Convert the NFA given in the following table to corresponding DFA.

(Consider Q0 as the initial state).

State

Input

a

b

Q0, **Q2**

Q3, Q1

01

02, 03

01, **00**

***02**

Q3, **Q2**

Q4

Q3

Q0, **Q1**, **Q2**

***Q4**

Q0

00, 03, 04 Q4, Q2

Q4.

(b) Prove that following language is not regular:

$\{a^i b^j \mid i \geq 1\}$

(c) Construct Pushdown Automata for grammar:

$S \rightarrow aSbb \mid aab.$

Attempt any two parts of the following:

(a) Construct Pushdown Automata that recognize the language:

$L = \{wewwe [a, b]^*\}$

(10 x 2 = 20)

(b) Construct a Turing Machine which works as a copying machine for $W \in (a, b)^*$.

(c) Design a Turing Machine for language:

$L = \{a^n b^{2n} \mid n > 0\}$

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

(a) Design a Turing Machine which can accept a language that consist of all strings with an

even number of a's and even number of b's.

(b) Explain Church's Thesis and Halting problem.

(c) Write a short note on following:

i. Universal Turing Machine

ii. Decidability and Un-decidability

iii. Post's Correspondence Problem (PCP)

iv. Recursive function theory

v. Generalized Transition graph

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BEET-404/BITT-404

Roll No.

Even Semester Examination, 2021-22 **Course Name : B.TECH**

Branch: CS

Semester: IV

Time: 3 Hours

Subject: Computer Organization & Architecture

Note: - Attempt all questions: All Questions carry equal marks

Q1. Attempt any four parts of the following

- (a) What do you mean by buses? Explain three types of buses? (b) Explain synchronous & asynchronous I/O techniques? (c) Explain load and store operations?
- (d) What is instruction cycle? Explain its phases?
- (e) Explain the terms instruction type and instruction format?

Q 2. Attempt any four parts of the following

- (a) Discuss various types of pipeline hazards? (b) What do you mean by polling explain it?

Max Marks: 100

Number of Printed pages: 1

(5 x 4 = 20)

(5 x 4 = 20)

- (c) What do you mean by RISC & CISC? Compare the architecture of RISC & CISC? (d) Explain multi bus organization?
- (e) What do you mean by direct & indirect addressing modes? Give example of each?

Q3. Attempt any two parts of the following

- (a) What is DMA? Explain it with proper diagram?

X (10 x 2 = 20)

- (b) What do you mean by interrupts? Explain how interrupts are handled in a system? (c) Explain the concept of virtual memory? How address translation is performed?

Q4.

Attempt any two parts of the following:

(10 x 2 = 20)

- (a) Explain general register organization & stack organization with proper diagram? (b) Construct a 32 bit ALU with proper diagram? Explain how does it work?
- (c) Explain what is horizontal & vertical micro programmed control unit? Write **down** its advantages & disadvantages?

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

- (a) What do you mean by addressing modes? Explain all types of addressing modes **with** proper examples?
- (b) What does memory hierarchy represent? Explain common framework of **memory** hierarchy?
- (c) How the concept of pipeline is implemented effectively in a computer system? **Write down** the advantages **of** pipeline

R-416

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BECT-402/BITT-402

Roll No.

Time: 3 Hours

Even Semester Examination, 2021-22

Course Name : B.TECH Branch: CS

Semester: IV

Subject: Database Management Systems

Note: - Attempt all questions: All Questions carry equal marks

Q 1. Attempt any four parts of the following

- (a) Example DBMS with its architecture
- (b) Define DDL and DML in context with DBMS
- (c) Explain Data independence with its types.
- (d) What are the different types of integrity constraints?
- (e) Define E/R Model with following
 - i. Entities and Relationships
 - ii. Attributes and different types of attributes

Q 2. Attempt any four parts of the following

- (a) Discuss Relational Algebra with suitable example. (b) Explain Tuple and Domain Relational Calculus (c) Define Normalization and its types with relevant examples (d) What do you understand by join statement? Explain (e) Explain the Query Processing Techniques.

Q3. Attempt any two parts of the following

- (a) List various locking techniques for concurrency control. (b) What is transaction? Explain ACID Properties with diagram. (c) Explain concurrency control based on time-stamp ordering.

Q4.

Attempt any two parts of the following:

Max Marks: 100

Number of Printed pages: 1

(5 x 4 = 20)

(5 x 4 = 20)

(10 x 2 = 20)

(10 x 2 = 20)

(a) Discuss SQL Injection Vulnerabilities detection methods by mentioning identification

tools.

(b) Compare and contrast between access controls models in DBMS with examples.

(c) Discuss the detection method of Intrusion Detection System in DBMS and Compare the

IDS with firewall.

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

(a) Explain Object Oriented and Object Relational Database with examples.

(b) What are distributed database? Explain with its importance.

(c) Illustrate data warehousing and data mining in DBMS

R-534

BECT-403/BITT-403

Roll No.

Time: 3 Hours

Even Semester Examination, 2021-22

Course Name : B.TECH

Branch: CSE

Semester: IV

Subject: Software Engineering

Note: - Attempt all questions: All Questions carry equal marks

Q 1.

Attempt any four parts of the following

Max Marks: 100

Number of Printed pages: 2

(5 x 4 = 20)

(a) Define the term software. Describe its various characteristics. (b) What is the Flow Chart? How is the flow charting technique useful for software development? (c) Define software crisis. What are the possible solutions for present software crisis?

(d) Explain Software Development Life Cycle. Discuss various activities during SDLC. (e) Define the following:

(i) (ii)

Aspect Oriented Model

Agile Model.

Q 2. Attempt any four parts of the following

(5 x 4 = 20)

(a) Describe the role of the following in the successful completion of a project

(a) Change Control

(b) Reviews

(b) What do you understand by software project estimation? Discuss various techniques used for software project

estimation

(c) Draw the complete DFD up to level 2 for Library Management system.

(d) Given the following values, compute function point when all complexity adjustment factor (CAF) and

weighting factors are average.

User Input = 50

User Output = 40

User Inquiries = 35

User Files = 6

External Interface = 4

(e) Discuss Various measures are used in project size estimation in detail.

Q3. Attempt any two parts of the following

(10 x 2 = 20)

(a) What are the input and output of system design? Describe of Different type of system

design.

(b) What is structured chart? Discuss Objectives of Using Structure Flowcharts. Describe

factors affecting system complexity in detail.

(c) What is the difference between the STLC (Software Testing Life Cycle) and SDLC (Software Development Life Cycle)? What is white box testing and explain in brief all types of white box testing?

R-535

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P.T.O.

Q4. Attempt any two parts of the following:

(10 x 2 = 20)

(a) What is software testing? Differentiate b/w SDET, Test engineer and developer.

(b) What is Test Plan and contents available in a Test Plan? What is the difference

between integration testing and system testing?

(c) Define the following:

(i)

(ii)

basis path testing

Verification and validation.

Q5. Attempt any two parts of the following:

(10 x 2 = 20)

(a) What is SQA? Describe three major methodologies for SQA in **detail**. (b) What is CMM? Describe different level of CMM in detail.

(c) Define the following:

(i) Effective modular design

(ii) ISO 9126 Standard.

R-535

2

BHUT-401

Roll No.

Even Semester Examination, 2021-22

Course Name : B.TECH

Semester: IV

Branch: Common to **All** Branches Subject: **Universal Human Values 2**

Time: **3** Hours

Note: - Attempt all questions: All Questions carry equal marks Q1. Attempt any two parts of the following

Max Marks: 50

Number of Printed pages: 2

(5 x 2 = 10)

- (a) Write a short note on the need for value education in professional courses.
- (b) Explain with example how skills and human values are complementary.
- (c) Self exploration is a process of dialogue between 'what you are' and 'what you really want to be'. Explain and illustrate.
- (d) What is the difference between animal consciousness and human consciousness?

Q 2. Attempt any two parts of the following

- (a) How can we ensure harmony in self ('I')?
- (b) Differentiate between the needs of self and the needs of body.
- (c) How can we ensure right utilization of the body?

(5 x 2 = 10)

- (d) "Shift from Pre-Conditioning to right- conditioning can be a solution of all conflicts."
Comments.

Q3. Attempt any one part of the following

- (a) What is justice? How does it lead to mutual happiness?

(10 x 1 = 10)

(b) What is the foundation value of relationship and how it is differ from complete value?

(c) Differentiate between intention and competence with examples.

PTO

Q4.

Attempt **any one part** of the **following**:

(10 x 1 = 10)

(a) What are the four orders of nature? Briefly explain them.

(b) What are the natural characteristics (swabhava) of human order? Explain.

(c) Define harmony in nature and how will you create it. Explain with examples.

Q5. Attempt **any** one part of the following:

(10 x 1 =10)

(a) What are the programs needed to achieve the comprehensive human goal? List and define each briefly.

(b) What do you understand by competence in professional ethics? Give two examples of its implications in industry.

(c) What are the implications of value based living at all four levels of living? Explain.

4

Attempt any two parts of the following: **10×2=20** (a) Construct nondeterministic push down automata that accept the following languages = $\{ab\}$ and $\{a^n b^n : n \geq 0\}$

on ==

(i)

$a^n b^n$

$L = \{a^n b^{2n} : n \geq 0\}$

(ii) $L = \{w^n : n \geq 1\}$

a

(b) Show that the language:

$L = \{0^n 1^n : n \geq 1\} \cup \{0^n 1^{2n} : n \geq 1\}$

is a context free language that is not accepted by any deterministic push down automata. (c) Construct a non deterministic push down automata that accepts the language generated by the grammar, $S \rightarrow aSb \mid aab$.

LO

5

Attempt any two parts of the following: **10×2=20** (a) For an alphabet $\{a, b\}$, consider a language L that consists of all strings with an even number

of a's and even number of b's. Design a turing machine that accepts the language L.

(b) Prove that if L is a recursive language then

L^c is also recursive.

Assume

Write short notes on any **two** of the following:

1. Turing Machine

(c)

(i)

(ii)

DS-1071]

Universal Turing

machine

Nuw vistu response Pro

recursive function theory

BHA

OA - 8

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[[2340]

DS-1 0 7 1 *

PAPER ID : 1071

TCS-403

Printed Pages: 4

Paper **ID** and Roll **No.** to be filled in your Answer Book

Roll **No.**

B. Tech.

(SEM. IV) EXAMINATION, 2012

THEORY OF AUTOMATA

Time: **3**

Hours]

Note :

1

[Total Marks :

Attempt all questions as per directions given there of. All questions carry equal marks.
Be precise in your answer.

Attempt any four of the following:

5×4=20

(a) What is a generalized transition graph ? Explain with a suitable example.

(b) Explain Chomsky hierarchy by taking suitable examples.

Consider the language s^* , where $S = \{xx, xxx\}$. In how many different ways can x^{23} be written as the product of words in S ? This means : How many different factorizations are there of x^{28} into **xx and xxx** ?

as **te**

Jer (d) Give recursive definitions for the following languages over the alphabet $= \{a,b\}$

DIO 201

bas

(e)

DS-1071]

(i)

The language EVENSTRING of all words of even length.

(ii) The language ODDSTRING of all words of odd length.

Show that any language that can be accepted by a transition graph can be accepted by a transition graph with an even number of

states.

1

[Contd...

(f) Describe in English the languages accepted by the following finite automater.

b

20

(i)

ab

9b

STOS MOITAMIMAXE (VM32) ab

(ii) ANTUQ, b vok a

БУБЕИ ID

08-2(e) Describe some applications of finite Automata theory in real life. s)
Minimize the states in the dfa depicted in the following diagram.

esgs (f)

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24

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(21)

22

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2

a, b

TatuoH &: emit

5×4-20

Attempt any four of the following:

(a) Convert to a DFA the following NFA :

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doorg

$\rightarrow P \{p, r\} \{q\}$

eldetine gift $\{r, s\} \{P\}$

*

$\{p, s\} \{r\}$

od mielo (d) asiqmuske

$\{g, r\} o$

on

wor

ats (b) Give a DFA accepting the language over to erant or no Erot

alphabet = $\{0,1\}$ as defined below:

Saiwollot the set of all strings beginning with a 1 that,

when interpreted as a binary integer, is a

dle of S. For example, strings 101, 1010, ebrow lls 0, 100 and 111 and 1111 are in the 1

(c)

(d)

to od

DS-1071]

are not.

Differentiate between Moore machine and Mealy machine by taking a suitable example.

Design NFA to recognize the following sets of String: abc , abd , and string: abc , abd , and $aacd$. Assume the alphabet is $\{a,b,c,d\}$

2

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[Contd...

1

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(95)

T=

0,1

betqsoos ton et tant egsugnaleert jxsino s'at

3 sts Attempt any two parts of the following: $10 \times 2 = 20$ twob(a) Prove that the following are not regular betarenog languages it atqooos tadi stamotus

(i) $\{n \mid n \text{ is a perfect square}\}$

98-801 (ii) $\{a^i b^j \mid i \geq j\}$

OSHA

JA

(b) Find context-free grammars for the following languages:

(i) $L = \{a^m b^n : n < m+3, n > 0, m \geq 0\}$ & (ii) $L = \{w \in (a+b)^* : \#a(w) \neq \#b(w)\}$

rent

(c) (i)

Show that the following grammar is ambiguous.

SAB aaB

Jai oe

(0)

garwollot of towar Aa 29ton trode 9thW

DS-1071]

A al

Blogabentur Iserevinu

→

(ii) Convert the grammar with

doqui evre9% (0)

productions S

$S \rightarrow ABa$

Aaab

BAC

to Chomsky normal form

3

[Contd...

6 *

967 *

PAPER ID: 9967

TCS-302/TCS-305

Printed Pages: 7

Paper ID and Roll No. to be filled in your Answer Book

Roll No.

B. Tech.

(SEM. III) (ODD SEM.) (REG. & BACK) EXAMINATION, 2012-13

COMPUTER BASED NUMERICAL & STATISTICAL TECHNIQUES (CBNT)

Time: **3 Hours]**

[Total Marks: **100**

UNIT - I

Answer any 4 question:

1.

$4 \times 5 = 20$

5

Find the root of the equation $f(x) = x^3 - 5x + 1$ using bisection method perform five iterations.

2.

Use Muller's method to find the root of

$\frac{1}{x}$

$\frac{1}{x}$

9967]

$f(x) = 4x^2 - \exp(x)$. Start with $p_0 = 4.0$, $p_1 = 4.1$ P_0

and $p_2 = 4.2$ and find P_3 , P_4 , P_5 .

3.

(Linear convergence at a double root)

P1

Start with $p_0 = 1.2$ and use Newton-Raphson

Po

iteration to find the double root $p = 1$ of the polynomial $f(x) = x^3 - 3x + 2$.

4. Differentiate between algebraic and

5.

transcendental equations.

What will happen if the bisection method is used with the function $f(x) = \tan(x)$ (x is radians) and the interval is $[3, 4]$ and $[1, 3]$.

1

5

5

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5

[Contd...

56.

Consider the function $f(x) = \cos(x)$.

3+2

(a)

Find the Newton raphson formula $g(P_{k-1})$

9967]

(b) We want to find the root $p = 3/2$ can we use $P_0 = 3$? Why?

UNIT - III

Answer any 2 question:

2×10=20

UNIT II

1

1.

A)

Evaluate

$\int_0^1 x \ln x \, dx$ Using trapezoidal 5+5

2

$1+x$

Answer any 2 question:

2×10=20

rule. Take $h = 0.125$.

1.

Using Newton's forward interpolation formulae, find a cubic polynomial which takes the following set of values:

10

B)

Given $y' = -y$ and $y(0) = 1$, determine the value of $y(0.01)$ by Euler's method.

X 0

1

2

3

2.

$f(x)$

1

2

1

10

2.

Solve $5x - y + z = 10$; $2x + 4y = 12$;

10

and $x + y + 5z = -1$ Using Gauss-seidial method.

3.

Let $f(x) = x +$

$2x$

(a)

3.

5+5

Verify that Simpson's rule ($M = 1$, $h = 1$) is exact for polynomials of degree ≤ 3 of the form

$f(x) = c_3x^3 + c_2x^2 + c_1x + c_0$, over $[0, 2]$.

Use the Taylor expansions for $f(x + h)$,

-

$f(x - h)$, $f(x + 2h)$ and $f(x - 2h)$ and derive the central - difference formula.

10

10

Use quadratic Lagrange interpolation based on the nodes $x_0 = 1$, $x_1 = 2$ and $x_2 = 2.5$ to approximate $f(1.5)$ and $f(1.2)$.

(b) Use cubic Lagrange interpolation based on the nodes $x_0 = 0.5$, $x_1 = 1$ and $x_2 = 2$,

x_3

$= 2.5$ to approximate $f(1.5)$ and $f(1.2)$.

2

$f(4)(X) = 0$.

$f(x+2h) - 4f(x+h) + 6f(x) - 4f(x-h) + f(x-2h)$

12

h

[Contd...

3

[Contd...

9967]

9967]

UNIT - IV

Answer any 2 question :

2×10=20

1.

Find the least square parabolic fit

10

2.

$$y = Ax^2 + Bx + C.$$

(a)

X_k

Y_k

-3

15

-1

bar 5

1

1

3

5

(b)

X_k

-3

Y_k

-1

1

7 22

25

25

1

3

Find the parabolic terminated cubic spline through (0, 0.0), (1, 0.5), (2, 2.0) and (3, 1.5) with the second derivative boundary conditions $S''(0) = -0.3$ and $S''(3) = 3.3$.

4

10

[Contd...ke

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3.

An economist wants to estimate the relationship 10 in a small community between a family's annual income and the amount that the family saves. The als following data from nine families are obtained: **Annual Income**

Annual Savings

(thousand of dollars) (Thousands of dollars)

12

0.0

13

0.1

14

0.2

15

0.2

16

0.5

17

0.5

18

0.6

19

0.7

20

0.8

Calculate **the** least-square regression line,

where annual saving is the dependent variable and annual income is the independent variable, and

interpret your results.

5

[Contd...

UNIT V

od Answer any 2 question:

2×10=20

1.

a)

Use the newton divide difference formula

5

to calculate $f(3)$, $f'(3)$ and $f''(3)$ from the following table:

X:

0

1

2

4

5

6

F(x): 1

14

15

5

6

19

b)

(Quadratic convergence at a simple root) **5**

Start with $P_0 = -2.4$ and use Newton-

Raphson iteration to find the root $p = -2$

of the polynomial $f(x) = x^3 - 3x + 2$.

2.

a)

Use Muller's method to find the root of

$\sqrt{x}$.

9967]

b)

3. a)

$f(x) = 1 + 2x - \tan(x)$. Start with $p_0 = 1.5$, P_0

=

$P_1 = 1.4$ and $P_2 = 1.3$ and find P_3, P_4, P_5 .

What are the difference between the

bisection method and the regula falsi method.

5

Start with $P_0 = -2.6$ and with $p_1 = -2.4$ **5** P_1

and use the secant method to find the root

$p = -2$ of the polynomial $f(x) = x^3 - 3x + 2$.

6

b)

Let $f(x) = x^3 - x + 2$.

i.

Find the Newton Raphson formula

$g(P_{k-1})$.

ii.

Start with P_0

=

-1.5 and P_1, P_2, P_3 .

2+3

[Contd...

9967]

7

[6600]

0\$ (b) Discuss the various possible hazards between read and write operations in an instruction pipeline and state the mechanism to detect and avoid these hazards. Also discuss the dynamic instruction scheduling.

(c)

the execution

Consider a program of 15,000 instructions by a linear pipeline processor with a clock rate of 25 MHz. Assume that the instruction pipeline has five stages and that one instruction per cycle and out-of-sequence executions are ignored. Setas (1)

Calculate the speedup factor in using this pipeline to execute the program as compared with the use of an equivalent nonpipelined processor with an equal amount of flow-

through delay.

a

edT (i)

dans

doold

What are the efficiency and throughput of this pipelined processor?

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blog CSOW bas 792

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PAPER ID: 1067

TCS-401/TIT-401 aminobi od prise **Printed Pages: 4**

Paper **ID** and Roll No. to **be filled in your Answer Book**

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(SEM. IV) EXAMINATION, 2012

COMPUTER ORGANIZATION

Time: **3 Hours]**

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Note :

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Attempt any Four parts of the following:

(a)

(8)

temot nodque [Total Marks: **100**

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Attempt all questions. All question carry equal marks. Vbreib 29g6isvosaib bus 2528ins
his 221021

(9)

$5 \times 4 = 20$

What do you mean by inter-register transfer? Discuss Bus transfer.

05-5 (b) Design Arithmetic Logic Shift unit that will perform different arithmetic, logic an
shift operation..

(c) Draw a diagram of bus system for four registers U geof 4-bits each. The bus is to be
constructed with multiplexers.bs

(d) 15 Show the hardware implementation for the noir following statements. The
registers are 4-bits in 2 aidlengths: gli odsto autste enols basta to TO A RO

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(woftevo) V (ngia)

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RI

R1

Isroitibros disq

T2

A ← R2

od

edyd bonoqqe od so

T3

AR3

(e)

Show the multiplication process using Booth's algorithm when the following binary numbers are multiplied: $(12) \times (-18)$

DS-1067]

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[5060]

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[Contd...

2

(f)

Represent the decimal number -0.654 into IEEE floating point format.

Attempt any Four parts of the following

$10 \times 2 = 201$

(a) Explain the Fetch and Execute cycle during execution of an instruction.

(b) Describe briefly a Hardwired Control.

(c)

Describe briefly

Hardwired

What is microinstruction? Describe the 00hs microinstruction format.

[addH & em

(d) What do you mean by the term Microprogram

(e) $05=4 \times 2$

3

Sequence' Explain the role played by it. us the Discuss advantages and disadvantages of Microprogrammed Control

510 M

(f)ten Explain wide-branch addressing and emulation.

Attempt any Two parts of the following

funans digo

$10 \times 2 = 20$

(a) Write a program to evaluate the arithmetic Hhiw expression: $X = (A + B * C) / (D + E * F / G + H)$ Using Three, Two, One and Zero address Instructions. (b) Design an efficient logic circuit shared by different netic branch conditions derived from the combination or stand alone status of the Flag register bits S (sign), V (overflow), C (End carry), Z (all O's). Specify the conditional branch instructions which can be supported by the circuit you have designed.

(c) Describe DMA with suitable block diagram. Why 2does DMA have priority over the CPU when 915 both request a memory transfer? Explain,

DS-1067]

2

4

Attempt any Two parts of the following (a)

000

$10 \times 2 = 20$

The access time of a cache memory is 100 ns and that of main memory 1000 ns. It is estimated that 80 percent, of the memory requests are for read and the remaining' 20 percent for write. The ((5) hit ratio for read accesses only is 0.9. mono(i) ani What is the average access time of the notourani system considering only memory read cycles? (ii) What is the average access time of the girls gnieu system for both read and write requests?

(11)

bensqmoos mis190x 90

What is the hit ratio taking into consideration

bmit() What is t

woll do the write cycles? to

(b)

A block set associative cache consists of a total

to tu of 64 blocks divided into four block sets. The

main memory containing 4096 blocks each consisting of 128 words.

How many bits are there in the main memory address?

(ii)

How many bits are there in each of TAG, SET and WORD field?

(c)

5

5

Discuss the concept and implementation of virtual memory. Also describe a suitable scheme for translation from logical address to physical address.

Attempt any Two parts of the following:

(a)

10×2=20

Give various architectural classification schemes. Also discuss the Flynn's and Feng's classification in detail.

[Contd...

DS-1067]

5

Attempt any two parts of the following:

(10 × 2 = 20)

Isnot (a) Explain **the** concept and working of locking techniques in concurrency control. What benefit does rigorous two-phase locking provide? How does it compare with older forms of two-phase locking? snitsa

sesdatberphase

od (b) no What do you mean by timestamps? Discuss timestamp- smena no ordering protocol in details. Also, discuss Thomas' write rule and multiple granularities in short. A)

(c) What do you mean by log-based recovery? Explain deferred and immediate modification versions of the

oalA Syone log-based recovery schemes. Also, compare these in terms of **ease** of implementation and overhead cost. hoitaviseng yonsbasqab **bas** nioj zaslaaol aaroela noftizogrooobs to asinsgong

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(d)

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Printed Pages: 4

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Paper ID and Roll No. to be filled in your Answer Book

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PAPER ID : 1068

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obute ai #2 (noitos)

B. Tech.

ЛОЛИЯ

200 al (SEM. IV) EXAMINATION, 2012

parv **DATA BASE MAN DATA BASE MANAGEMENT SYSTEM**

Time: **3 Hours]**

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42)

[Total Marks: **100**

2010 (1)

Attempt ALL questions.oup svið All Questions carry equal marks.

Note olso (1)

1st gao() in case **of** numerical problems assume data

1

wherever not provided.

Be precise in your answer.

no nisdien(IV) Be precise

927

ori Attempt any four parts of the following: (a) Define the following terms:

(1)

Data redundancy and consistency

ednum Referential Integrity

(11)

di

noftib.(iii) Data atomicity

$(5 \times 4 = 20)$

nes ab

woy objedW

mon

(iv) Domain Constraint no

to 101stego(v)

Data Models mortsole or ansai

Data Models

(b) Explain the concept of keys in DBMS. Also describe anoitsisqo the different types of keys in details.

(b)

(c) Draw an E-R diagram for a hospital with a set of avi sausai patients and a set of medical doctors, with each patient a of the various co a of the various conducted tests is also associated.

(d) Explain two-tier and three-tier architecture of DBMS in details.

(e)

[3040]

DS-1068]

vingsini istings A

What do you mean database language and interfaces? What is the difference between DDL and DML? Explain.

18001-201

[Contd...

2

(f)

Discuss the candidate key, primary key, super key, composite key and alternate key.

Xoog enA woy n

(a)

ID 10e8

Attempt any four parts of the following: of the following: (5 x 4 = 20) Given the relational schema: ENROLL (S#, C#, Section), S# is student number. TEACH (Prof, C#, Section), C# is course number. MET ADVISE (Prof, S#), Prof is thesis advisor of S#. PRE-REQ (C#, pre-C#), pre-C# is prerequisite course. GRADE (S#, C#, grade, year)

Taucht & SemiT STUDENT (S#, Sname), Sname is student name. Give queries expressed in SQL and tuple calculus. (1) List of students taking course with smith or jones. Bab smua(ii) am List all students taking at least one course that their advisor teaches,

00: edhe

(OS

w

List those professors who teach more than one section of the same course/

(iv) List all student number and course number who

got grade A.

(v) List all students number and course number. (b) What do What do you mean by Triggers? How they are different from Assertions, discuss. nismo

Discuss the selection, projection and join operator of

odinoaab oe relational algebra with suitable example. (0)

(d) Discuss the various types of JOIN operations.

(5).

to 192 Differentiate OUTER and INNER joins. Insin(e) on What are the characteristics of SQL? Discuss five basic aggregate functions with a suitable example.

2MB (1) To Write short notes on following:

(i) Referential integrity

nisixd (b)

elistab ni

290shisini (ii) Domain calculus som uoy ob tedW

IMG bred shib or ai tadW

DS-1068]

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2

3

Our parts of

30s Attempt any the following: (**5** x 4 = 20) Write and prove Armstrong's Axioms for functional

25upi (a)

- 10 dependencies. Wood v

-owl au

Tobio (b)

What is th

What **is** the significance **of** normalization **of the database**? Define 1NF and 2NF.

and 2NF.asdq-ows to ammot

Corus (c) mis **What** do you mean by canonical **cover**? **Consider the** sti asmo following **set** F of functional dependencies **on schema** (A, B, C): asithelimsig elqitlum bas slui

nisixal SyABC, BC, AB, AB → Co) on to anol Compute **the** canonical cover of F.nstab ni
sac(d)

4

What do you mean by multi-valued dependency? Also, 3200 beer explain **the** concept of 4NF in short.mo

(e)

(f)

Discuss lossless join and **dependency preservation** properties of a decomposition.

Consider the following two **sets** of **functional** dependencies:

FAC, $AC \rightarrow D$, $E \rightarrow AD$, $E \rightarrow H$ and

G = ACD, $E \rightarrow AH$

Check, whether F and G **are equivalent**.

Attempt any two **parts** of **the** following:

(10 × 2 = 20) (a) Explain the ACID properties of **transaction**. **Draw** and explain **the state** diagram of a transaction. **What are the** different anomalies **that** may occur due to **concurrent** execution?

(b)

What do you mean by schedule in **the context** of concurrent execution of transactions in **RDBMS**? **What is** serializable schedule? Discuss **the** various **types of** serializability with a suitable example.

(c) **What** do you mean by deadlock? When **are the** various conditions under which **deadlock** occur? **Discuss the** wait-die and wound-wait in **detail**.

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[Contd...

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PAPER ID: 1065

TCS-303

Printed Pages: 3

Paper ID and Roll No. to be filled in your Answer Book Roll No.

B. Tech. d'W

(SEM. III) (ODD SEM.) (REG. & BACK) EXAMINATION, 2012-13

DATA STRUCTURE

Time: **3 Hours]**

[Total Marks: **100**

UNIT - I

1

Do any **four** :

4×5=20

(a)

00

(c)

1065]

What do you understand by space and time complexity of an algorithm? Explain with example.

(b) How will you decide whether a given data structure are linear or non linear, homogeneous or heterogeneous? Explain with example.

What is dynamic memory allocation? Differentiate between malloc() and calloc().

(d) Write a program to shift all zero elements in an array toward end of the array.

(e) What are the various ways to represent a stack in memory? Discuss the relative merits and demerits of each.

(f)

What is recursion? How it is different from iteration? Write a recursive function to find the GCD of two integers.

1

[Contd...

UNIT - II

UNIT - IV

2

Do any **two**:

10×2=20

4

Do any **two** :

(a)

What do you understand by linked list? What are its advantages and disadvantages over an array? ` Write a program to reverse a singly linked list.

10

(a)

How searching for a number in sorted list affects the execution time? How binary search is different from linear search? Write a function for binary search.

10×2=20

10

(b) How can we perform insertion and deletion in D-eque, represented as D-eque? Write enqueue and dequeue functions for D-eque.

10

(b)

(c)

What is circular queue? How it is different 10 from linear queue? Give the linked list representation

(c)

of circular queue with its underflow and overflow check.

What is hashing? How it is different from linear and binary search? What is collision? Can it be eliminated completely? Explain with example. How insertion sort is different from selection sort? Which sorting algorithm will work best if the list is already sorted, explain and write the C function as well.

10.

10

UNIT - III

UNIT - V

3

Do any **two**:

(a)

$10 \times 2 = 20$

5

Do any **two** :

(a)

$10 \times 2 = 20$

10

1065]

What is Binary Search Tree? Explain with an 10 example. Consider the postorder and inorder

en given for a binary tree:

Postorder: 4,11,5,3,8,5,10,9,2,1

Inorder: 4,3,11,5,1,8,2,17,10,9

What is indexing? Explain the primary, clustering and secondary index suitable example.

(b) What is text file? How it is different from other 10 types of **files**. List the various advantages and disadvantages of sequential. Direct and indexed file organization.

Find the binary tree.

(b)

What are the disadvantages of using an

10

(c)

Write short notes on following with example :

10

array for representation of binary tree? Explain.

(i)

B+ Tree Index Files

Write a function to find the number of nodes in a binary tree.

(ii)

B Tree Index Files.

(c)

10

What is the threaded binary tree? How it can be represented in computer memory?

Write a function to search an element in a binary search tree.

2

[Contd...

1065]

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[3120]

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*DS-1050-10 6 3 *

PAPER ID: 1050-1063

TCS-402/TIT-402

Printed Pages: 3

Paper ID and Roll No. to be filled in your Answer Book

Roll No.

B. Tech.

(SEM. IV) EXAMINATION, 2012 **UNIX & SHELL PROGRAMMING**

bis grow baqgpia as toning 100Y

WoTime: : **3** Hours]mbab gning order

Attempt ALL questions.

[Total Marks: **100**

Note:

(ii)

All Questions carry equal marks.

(1)

In case of numerical problems assume data wherever not provided.

(iv) Be precise in your answer.

1

Attempt any FOUR parts of the following:

(a)

(5×4= 20)

Describe how UNIX is more secured than other operating systems. sd bas smart 1920

di

Set

brunego(b) 99How many types of editors we can use in UNIX operating system. What are the main functions of vi editors.

[6001-0201-20

iods bng (C)

Explain the purpose of include files in UNIX.

including

(d)

Explain the role of File descriptors, File table and Incode table in Unix File System.

(e)

9010

2

Draw the block diagram of Unix system kernel. Explain di various components.poz odt
taill

191

ni bagno Attempt any FOUR parts of the following: (5×4= 20)

(a)

Write a short noted on internal and external commands of UNIX.

...b **DS-1050-1063]**

1

12301-020[Contd...

S: Bege (b) in What does kill \$! Do? Explain with an example. loo(c) werExplain the task of following commands:

(111)

(v)

tty

(ii)

cat

(iv)

gzip

ФИ Под banner

cmp \$7 8

(e)

You are editing a file in vi, and want to make a temporary exit to the shell. How will you do that and

ОиIM then come back to vi again?U

(f)

Your printer has stopped working, and you want to 00: ahsM isbe sure whether the printing daemon is working. How

этой

(5×4= 20)

Determine what the built-in shell variables are for your

account. Which ones are exported?

will you ensure that?

ПА

3

Attempt any

four parts of the following:

stab: 9mm(a)

(b)

Explain:600

(vi)

(i) Shell metacharacters

(OS =AX2)

Shell variables qvs

A

edio (c)

(b) (i)

(ii)

What is the difference between shutdown and halt.

Explain mounting of a file system in UNIX. When is unmounting a file system not possible? (c) Explain the utility provided by the UNIX for managing and tracking to files. Suppose we want to check a C program for syntax and data type errors. Which UNIX utility can be used to display messages that point out, possible problems? Explain the procedures to use this utility.

Attempt any two parts of the following: (10 × 2 = 20) Write short note on language development tools YACC, LEX, M4 in UNIX.

5

(a)

(b)

(c)

Describe the following tools troff, nroff, tbi, Dbx, and Adb

Describe Library functions and system calls with an example.

Write a shell script that prints the current date, your user name and the name of your login shell.

XMU ni(d) Write a shell script that creates three background

iv to anosomut nism

processes, waits for them all to complete, and then

display a simple message.

X(e)

(e)

I

exotibo

Explain the various types of UNIX shells and their

basic characteristics. to slo srit misiga

moteve slil xin ni sidelobosal

(b)

nis4 Attempt any two parts of the following: ($10 \times 2 = 20$)

(a) (1)

(05-x2)*

List the sequence of events that occur when one turn on the computer.

ebaminamos tum (1) What is the various security features offered in

sbDS-1050-1063]

UNIX? Explain any two in detail.

2

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020[Contd...

DS-1050-1063]

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[3220]

Sub Code: CST006 ROLL NO.....

ODD SEMESTER EXAMINATION, 2024 – 25

2nd Year (3rd Sem) B.Tech.

CYBER SECURITY

Duration: 2:00 hrs Max Marks: 50 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. a) Explain the importance of cybersecurity and its applications. b) What are Cyber threats? Name four types of cyber threats. c) What are the motives behind attackers conducting cyberattacks? Explain with examples.	5x2=10
Q 2.	Answer any two parts of the following. a) What is cyber forensics? Discuss its historical background. b) Explain the role of digital evidence in cyber forensics. Why is it crucial in investigations? c) Discuss the challenges in computer forensics.	5x2=10
Q 3.	Answer any two parts of the following. a) Discuss the security attacks on mobile/cell phones. b) Discuss the importance of authentication service security in mobile computing. c) Explain the proliferation of mobile and wireless devices.	5x2= 10
Q 4.	Answer any two parts of the following. a) Explain the impact of web threats on organizational security and privacy. b) Discuss the ethical dimension of cybercrimes. c) Explain cyber terrorism in detail with examples.	5x2= 10

Q 5.	Answer any two parts of the following. a) What is the importance of cyber security regulations and laws? b) Discuss the main objectives of India's national cyber security policy. c) Discuss the key offences under the Information Technology Act in India.	5x2= 10
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Sub Code: CST006 ROLL NO..... 0.70.207 | 21/06/2025 08:52:45

EVEN SEMESTER EXAMINATION, 2024 – 25

2nd Year (IVSem) B.Tech.

Cyber Security

Duration: 2:00 hrs Max Marks: 50 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	06_005 103.1 ⁷ :4 ⁵ Answer any two parts of the following. a) Define vulnerability, threat, and attack. Give suitable examples. b) Differentiate between active and passive attacks with examples. c) Design a basic security model for a small enterprise network.	5x2=10
Q 2.	CST ⁰ 1/06/2025 08:5 ² Answer any two parts of the following. a) What is cyber forensics? Explain its importance in cybercrime investigation. b) Discuss various challenges faced in forensic auditing. c) Describe the digital forensic investigation process with an example.	5x2=10
Q 3.	3.170.70.207 ² Answer any two parts of the following. a) Define mobile security and explain its challenges. b) Illustrate credit card frauds in mobile environments with real examples. c) Design organizational policy for handling mobile security.	5x2= 10
Q 4.	CST006_005 1 ⁰ Answer any two parts of the following. a) What are the major costs of cybercrime for organizations? b) Evaluate the implications of intellectual property theft in cyberspace. c) Explain the impact of social media threats on corporate data privacy.	5x2= 10 52:4 ⁵
Q 5.	21/06/2025 ⁰ Answer any two parts of the following. a) What are the key features of the IT Act 2000 and its 2008 amendments? b) Evaluate the Indian National Cyber Security Policy. c) Describe the role of electronic evidence in Indian legal systems.	8:5x2= 10

Sub Code: CST-006 ROLL NO..... ODD SEMESTER EXAMINATION,
2023 – 24

IInd yr , B.Tech

Cyber Security

Duration: 3:00 hrs Max Marks: 50 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. a) What is the difference between snooping and spoofing? b) Differentiate between active and passive attacks. c) What do you mean by confidentiality and explain its threats?	5x2=10
Q 2.	Answer any two parts of the following. a) Explain the need of cyber forensics and computer forensics. b) What are the special techniques for forensics auditing? c) Define the term: a) Cyber Terrorism b) Cyber Espionage	5x2=10
Q 3.	Answer any two parts of the following. a) Discuss the major security challenges faced by wireless technology? b) How message authentication code can be used to achieve message authentication and confidentiality? c) Explain digital signature and its authentication protocol.	5x2= 10
Q 4.	Answer any two parts of the following. a) Discuss the potential threat faced by any organization and how these web threat can be prevented. b) How can you trace the mindset of hackers & cyber criminals? Is there any strategy to trace the steps of a hacker. Justify your answer. c) Explain integrity and its threats too.	5x2= 10

Q 5.	Answer any two parts of the following. a) Briefly discuss some of the regulatory authorities for prevention of cyber-crime. b) What are the challenges and constraint faced by Internet Governance? c) What is the IT Act of 2008?	5x2= 10
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Sub Code: BCST303 ROLL NO.....

III SEMESTER EXAMINATION, 2023 – 24

IInd year B.Tech. – Computer Science & Engineering

DATA STRUCTURE

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Define algorithm and its characteristics and also define best case, average case and worst case for analyzing the complexity of a program. b) Give brief description about the priority queues. write an algorithm to implement priority queue c) List the various asymptotic notations? Explain Big Oh notations along with suitable diagram d) Apply suitable data structure to convert infix to postfix expression: $K + L - M * N + (O^P) * W / U / V * T + Q$ e) Given $A = [10, 30, 40, 80, 200, 700, 927, 1000, 1410, 1578]$ Here $N = 10$, Value = 6 Find the Time Complexity in terms of (a) Linear Search Algorithm (b) Binary Search Algorithm f) What is stack? Why it is known as LIFO? Write algorithm of PUSH and POP operation on stack	5x4=20
Q 2.	Answer any four parts of the following. a) The order of nodes of a binary tree in inorder and postorder traversal are as follows: In order : B, I, D, A, C, G, E, H, F. Post order: I, D, B, G, C, H, F, E, A. (i) Draw the corresponding binary tree. (ii) Write the pre order traversal of the same tree. b) Write a program in c language to sort the sequence 13, 11, 74, 37, 85, 39, 22, 56, 25 using Insertion sort c) Write an algorithm for Quick sort. Use Quick sort algorithm to sort the following elements: 2, 8, 7, 1, 3, 5, 6, 4 d) Discuss doubly linked list. Write an algorithm to insert a node after a given node in singly linked list. e) What is circular Queue? Write a C code to insert an element in circular queue? f) What is queue? Why it is known as FIFO? Write an	5x4=20

	algorithm to delete_last and delete_random an element from a simple queue.	
Q 3.	Answer any two parts of the following. a) Define an AVL tree. Obtain an AVL tree by inserting one integer at a time in the following sequence. 150, 155, 160, 115, 110, 140, 120, 145, 130, 147, 170, 180. Show all the steps. b) (i) What is Graph? Explain matrix and linked list representation of a graph. Also give the application of Graph (ii) Differentiate depth-first search and breadth-first search traversal of a graph with	10x2= 20

	suitable examples. c) Construct a B-Tree of order 5 with the following sequence of integer. 10,90,20,80,30,70,40,60,50,35,55,15,25,5,75,85,95,45,100,22,12	
Q 4.	Answer any two parts of the following. a) Explain ADT and Concrete data structure with examples? b) Write short notes on (i) Minimum Spanning Tree (ii) Hashing c) What is Bubble Sort? How it is different from Selection Sort? Explain how	10x2 = 20

	the following list can be sorted using the bubble sort algorithm: 13,7,9,32,76,96,100,22,88,6,17	
Q 5.	Answer any two parts of the following. a) What do you mean by min heap? Create a min heap for the following data: 10,30,16,12,25,30,14,2,7 After creation of min heap, perform one delete operation and represent final min heap b) Write an algorithm to reverse a single linked list c) Create a Binary Search Tree for the following data and do in-order, Pre-order and Post-order traversal of the tree. 50, 60, 25, 40, 30, 70, 35, 10, 55, 65, 5	10x2 = 20

Sub Code: BCST303 ROLL NO.....

III SEMESTER EXAMINATION, 2023 – 24

IInd year B.Tech. – Computer Science & Engineering

DATA STRUCTURE

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Define algorithm and its characteristics and also define best case, average case and worst case for analyzing the complexity of a program. b) Give brief description about the priority queues. write an algorithm to implement priority queue c) List the various asymptotic notations? Explain Big Oh notations along with suitable diagram d) Apply suitable data structure to convert infix to postfix expression: $K + L - M * N + (O \wedge P) * W / U / V * T + Q$ e) Given $A = [10, 30, 40, 80, 200, 700, 927, 1000, 1410, 1578]$ Here $N = 10$, Value = 6 Find the Time Complexity in terms of (a) Linear Search Algorithm (b) Binary Search Algorithm f) What is stack? Why it is known as LIFO? Write algorithm of PUSH and POP operation on stack	5x4=20
Q 2.	Answer any four parts of the following. a) The order of nodes of a binary tree in inorder and postorder traversal are as follows: In order : B, I, D, A, C, G, E, H, F. Post order: I, D, B, G, C, H, F, E, A. (i) Draw the corresponding binary tree. (ii) Write the pre order traversal of the same tree. b) Write a program in c language to sort the sequence 13, 11, 74, 37, 85, 39, 22, 56, 25 using Insertion sort c) Write an algorithm for Quick sort. Use Quick sort algorithm to sort the following elements: 2, 8, 7, 1, 3, 5, 6, 4	5x4=20

	d) Discuss doubly linked list. Write an algorithm to insert a node after a given node in singly linked list. e) What is circular Queue? Write a C code to insert an element in circular queue? f) What is queue? Why it is known as FIFO? Write an algorithm to delete_last and delete_random an element from a simple queue.	
Q 3.	Answer any two parts of the following. a) Define an AVL tree. Obtain an AVL tree by inserting one integer at a time in the following sequence. 150, 155, 160, 115, 110, 140, 120, 145, 130, 147, 170, 180. Show all the steps. b) (i) What is Graph? Explain matrix and linked list representation of a graph. Also give the application of Graph (ii) Differentiate depth-first search and breadth-first search traversal of a graph with	10x2= 20

	suitable examples. c) Construct a B-Tree of order 5 with the following sequence of integer. 10,90,20,80,30,70,40,60,50,35,55,15,25,5,75,85,95,45,100,22,12	
Q 4.	Answer any two parts of the following. a) Explain ADT and Concrete data structure with examples? b) Write short notes on (i) Minimum Spanning Tree (ii) Hashing c) What is Bubble Sort? How it is different from Selection Sort? Explain how	10x2 = 20

	the following list can be sorted using the bubble sort algorithm: 13,7,9,32,76,96,100,22,88,6,17	
Q 5.	Answer any two parts of the following. a) What do you mean by min heap? Create a min heap for the following data: 10,30,16,12,25,30,14,2,7 After creation of min heap, perform one delete operation and represent final min heap b) Write an algorithm to reverse a single linked list c) Create a Binary Search Tree for the following data and do in-order, Pre-order and Post-order traversal of the tree. 50, 60, 25, 40, 30, 70, 35, 10, 55, 65, 5	10x2 = 20

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B.TECH. (SEMESTER-III) HASSO

sepsinsyosabbne 200sinsvbs **DATA STRUCTURE**

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Time: 03:00 Hours

Max Marks: 100

1.

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Attempt **any four**:

[5×4=20]

(a)

What is the data structure? Differentiate between data type and data structures.

(b)

Convert the infix $(a+b)*(c+d)/f$ into postfix & prefix expression

(c)

Define Doubly Linked List. Write a program to print out the elements of a singly Od nedonu linked list.

(d)

Write a program to evaluate a postfix expression.

(e)

Explain the procedure to insert a new node in the (a) Beginning (b) End of the list

2.

Who Attempt **any four**:

(a)

Jell be (b)

[5×4=20]

Explain the following algorithm of a circular linked list. (a) inset the node at the beginning (b) delete a node from beginning

What do you mean by Array? Describe the storage structure of Array. Also

Explain Various types of Array in detail.

Draw a Binary search tree for the following input list 60, 25, 75, 15, 50, 66, 33,

44. Trace the algorithm to delete the nodes 25,75,44 from the tree.

(d)

Construct an expression tree for the input $ab+cde+^{**}$

(e)

What are Circular Queue and Priority Queue? Write an algorithm to insert and delete an element from a Circular Queue.

TCS-303/TIT-303/2420

(1)

[P.T.O.]

3.

4.

5.

Attempt **any four**:

(a)

(b)

(c)

RU1474 [10×2=20]

Explain Inorder, Preorder and Postorder Traversal operation on Binary tree with example. Construct a tree for the given inorder and postorder traversals. Inorder:

DGBAHEICF Postorder: GDBHIEFCA

State different File Organizations and discuss the advantages and disadvantages of each of them. Write a short note on indexed file organization.

Convert following Infix expression into Postfix expression using Tabular method.

$a-b/c * d+e*f/g$

Attempt **any four**:

(a)

(b)

(c)

[10×2=20]

Write and test a program that performs the operation Insert, Delete Min, Build Heap, Findmin, DecreaseKey, Delete, and Increase Key in a binary Heap

=

Given input { 4371,1323,6173,4199,4344,9679,1989} and a hash function $h(X) = X \pmod{10}$. Show the resulting 1.Separate chaining table 2.Open addressing hash table using linear probing 3.Open addressing hash table using Quadratic probing 4.Open addressing hash table with second hash function $h_2(X) = 7 - (X \pmod{7})$

Explain Different types of Hash Function with example. Explain Radix Sort with the help of example.

Attempt **any four** :

(a)

[10×2=20]

Write an algorithm for inserting and deleting element in an doubly linked list.

Explain linear linked list implementation of stack and queue.

(b)

What do you mean by balanced trees? What are the categories of AVL rotations? What do you mean by balance factor of a node in AVL tree?

(c) What do you mean by separate chaining? Write the advantage and disadvantage of separate chaining. What do you mean by open addressing?

TCS-303/TIT-303/2420

(2)

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Sub Code: CST-003 ROLL NO.....

III SEMESTER EXAMINATION, 2023 – 24

IInd yr B.Tech. – (Civil Engg/CS&E/AI&ML/EE/E&EE/IT)

Data Structures and Algorithms

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Differentiate linear search and binary search. b) Define Time-Space trade-off.	5x4=20
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	c) Write an algorithm for Breadth First Search (BFS) traversal of a graph. d) Write advantages of AVL tree over Binary Search Tree (BST). e) Write short notes on min heap. f) Define best case, average case and worst case for analyzing the complexity of a program.	
Q 2.	Answer any four parts of the following. a) List the advantages of doubly linked list over single linked list. b) What do you mean by Threaded Binary Tree? c) Explain Heap sort with example? d) Write short note on Priority Queue. e) Write an iterative function to search a key in Binary Search Tree (BST). f) Write different representations of graphs in the memory.	5x4=20
Q 3.	Answer any two parts of the following. a) What is Hashing? Explain division method to compute the hash function and also explain the collision resolution strategies used in hashing. b) Write a C program to insert a node at kth position in single linked list. c) Write algorithms of insertion sort. Implement the same on the following numbers.- 13, 16, 10, 11, 4, 12, 6, 7. also calculate its time complexity	10x2= 20
Q 4.	Answer any two parts of the following. a) Insert the following sequence of elements into an AVL tree, starting with empty tree 71,41,91,56,60,30,40,80,50,55 also find the minimum array size to represent this tree. b) If the in order of a binary tree is B,I,D,A,C,G,E,H,F and its post order is I,D,B,G,C H,F,E,A then draw a corresponding binary tree with neat and clear steps from above assumption. c) What is circular Queue? Write a C code to insert an element in circular queue?	10x2= 20

Q 5.	Answer any two parts of the following. a) What do you mean by stack? Explain all stack operations with example? b) What is B-Tree? Write the various properties of B- Tree. Show the results of inserting the keys F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B in order into a empty B-Tree of order 5. c) How to represent the polynomial using linked list? Write a C program to add two polynomials using linked list.	10x2= 20
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Sub Code: CST-003 ROLL NO.....

III SEMESTER EXAMINATION, 2023 – 24

IIInd yr B.Tech. – (Civil Engg/CS&E/AI&ML/EE/E&EE/IT)

Data Structures and Algorithms

Duration: 3:00 hrs Max Marks: 100 *Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.*

Q 1.	Answer any four parts of the following. a) Differentiate linear search and binary search. b) Define Time-Space trade-off. c) Write an algorithm for Breadth First Search (BFS) traversal of a graph. d) Write advantages of AVL tree over Binary Search Tree (BST). e) Write short notes on min heap. f) Define best case, average case and worst case for analyzing the complexity of a program.	5x4=20
Q 2.	Answer any four parts of the following. a) List the advantages of doubly linked list over single linked list. b) What do you mean by Threaded Binary Tree? c) Explain Heap sort with example? d) Write short note on Priority Queue. e) Write an iterative function to search a key in Binary Search Tree (BST). f) Write different representations of graphs in the memory.	5x4=20
Q 3.	Answer any two parts of the following. a) What is Hashing? Explain division method to compute the hash function and also explain the collision resolution strategies used in hashing. b) Write a C program to insert a node at kth position in single linked list. c) Write algorithms of insertion sort. Implement the same on the following numbers.- 13, 16, 10, 11, 4, 12, 6, 7. also calculate its time complexity	10x2= 20
Q 4.	Answer any two parts of the following. a) Insert the following sequence of elements into an AVL tree, starting with empty tree 71,41,91,56,60,30,40,80,50,55 also find the minimum array size to represent this tree. b) If the in order of a binary tree is B,I,D,A,C,G,E,H,F and its post order is I,D,B,G,C H,F,E,A then draw a corresponding binary tree with neat and clear steps from above assumption.	10x2= 20

	c) What is circular Queue? Write a C code to insert an element in circular queue?	
Q 5.	Answer any two parts of the following. a) What do you mean by stack? Explain all stack operations with example? b) What is B-Tree? Write the various properties of B- Tree. Show the results of inserting the keys F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B in order into a empty B-Tree of order 5. c) How to represent the polynomial using linked list? Write a C program to add two polynomials using linked list.	10x2= 20

Sub Code: CST003 ROLL NO..... ODD SEMESTER EXAMINATION,
2024 - 25

2nd Year (III Sem) B.Tech.

Data Structures and Algorithms

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Define best case, average case and worst case for analyzing the complexity of a program. (ii) What condition will indicate that the queue is a) FULL b) EMPTY b) Draw all possible binary Tree with 3 nodes (10 marks) c) What is searching? Which searching technique is best and why? Write a Algorithm to perform Binary search.
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Write a function to check stfull and stempty condition in stack. (ii) Define row major and column major Array representation with suitable example. b) What is graph? Define simple graph, directed graph,cyclic and acyclic graphs. Explain the memory representation of graph and diagraph c) Define Hashing. How do collisions happen during hashing? Explain the different techniques resolving of collision.
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Which are the two standard ways of traversing a graph? (ii) Evaluate the following prefix expression " ++ 26 + - 1324" b) In quick sort how pivot is chosen? Give the complexity of quick sort and merge sort. c) Draw an AVL tree on following inputs. 55,65,70,75,80,90,100,110

Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) In what way an AVL tree is better than binary tree? (ii) What is the difference between a Stack and an Array? b) If the in-order traversal of a binary tree is B,I,D,A,C,G,E,H,F and its post order traversal is I,D,B,G,C,H,F,E,A . Determine the tree. c) Transform the following expression $(A+B) * C + D / (B + A * C) + D$ into equivalent postfix using stack
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Use selection sort algorithm to sort the following 15,22,30,10,64,1,3,9,52 (ii) Write a Function in 'C' to count the no of node in linked list. b) How will inorder, preorder and postorder traversals print the elements of a tree? c) Show the result of inserting the keys. F, S, Q, K, C, L, H, T, V, W, M, R, N , P, A, B, X, Y, D, Z, E in the order to an empty B-tree of degree-3.

Sub Code: CST 003 ROLL NO..... EVEN SEMESTER EXAMINATION,
2024 – 25

2nd Year (IV Sem) B.Tech.

Data Structures /Data Structures & Algorithms

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Explain any three basic operations that can be performed on data structures with examples. (ii) Differentiate asymptotic notations with their suitable graphs. b) Explain classification of data structure with diagram. c) Given a sorted array: <code>arr[] = {4, 10, 15, 18, 23, 31, 42, 55, 63, 78}</code>. Use Binary Search to find the index of element 31. Show each step of your search and state the number of comparisons made.
Q 2.	Answer any two parts of the following. UNIT-2 (10x2= 20) a) (i) Consider the following sequence of operations on an empty stack. <code>push(54); push(52); pop(); push(55); push(62); s = pop();</code> Consider the following sequence of operations on an empty queue. <code>enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();</code> Evaluate the value of <code>s + q</code>? (ii) Evaluate the following postfix expression: <code>6 2 3 + - 3 8 2 / + * 2 ^ 3 +</code>. b) Convert the following infix expression <code>A + (B * C (D / E F) * G) * H</code> into its equivalent postfix expression using stack table. c) Define and explain the concept of a Circular Queue and Compare the time complexity of insertion and deletion in a simple queue, circular queue, and priority queue.
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Write a code for inserting a node at the end of the singly linked list. (ii) Explain the advantages and disadvantages of list over array.

	b) Explain properties of Circular doubly Linked List with algorithm of deletion of a node from given location. c) Describe linked representation of stack and queue with suitable diagram.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) In a binary tree, the number of internal nodes of degree 1 is 5, and the number of internal nodes of degree 2 is 10. How many number of leaf nodes in the binary tree? (ii) Suppose the values 10, -4, 15, 30, 20, 5, 60, 19 are inserted in that order into an initially empty binary search tree. Let T be the resulting binary search tree. Construct the tree and evaluate the number of edges in the path from the node containing 19 to the root node. b) Construct AVL tree step by step for the following data 21,26,30,9,4,14,28,18,15,10,2,3,7. c) Define graph with its terminologies and types with diagrams.
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. If the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted? (ii) Write down the pseudo code for insertion sort with explanation. b) Explain hashing? Also describe collision resolution techniques with examples. c) Explain the algorithm for Merge sort and give a step-by-step solution for the given data elements. 39, 9, 81, 45, 90, 27, 72, 18

TCS-404

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Even Semester Examination, 2017-18

B.TECH. (SEMESTER-IV)

DATABASE MANAGEMENT SYSTEM

Time: 03:00 Hours

Note: (i) Attempt all question.

Max Marks: 100

(ii) All question carry equal marks.

Q1.

Attempt any **four** parts of following:

(5x4=20)

(a)

Explain three tier architecture of DBMS in details.

(b)

Explain the concept of mapping constraints.

(c)

What is the difference between specialization and generalization with respect to database?

(d)

Explain the distinctions among the terms primary key, candidate key and super key with suitable example.

(e)

Explain the difference between a weak and a strong entity set.

(f)

Q2.

Define E-R Diagram. Draw E-R diagram with Customer, Loan and Payment sets.

Attempt **any four** parts of following:

(5x4=20)

(a)

What is Relational Algebra? Define Relational Algebra Operation Cross product with example.

(b)

What is join? List and explain various types of joins.

(c)

What are triggers? Explain the advantages and the needs.

(d)

What are views? How is view defined? Explain with suitable example. What are the advantages of views?

(e)

What is a constraint in database? Explain types of constraints with a suitable

TCS-404/2120

example.

(1)

[P.T.O.]

(f) List and explain DML statements with suitable example?

Q3.

Attempt any **two** parts of following:

(a)

Q4.

(10x2=20)

What do you mean by normalization? List various normalization techniques.

Explain why 4NF is a normal form more desirable than BCNF.

(b)

(1) Consider the relation schema $R(A, B, C)$, which has the FDs $AB \Rightarrow C$, $C \Rightarrow A$,

identify the best normal form that R satisfies (1NF, 2NF, 3NF or BCNF).

(2)

What do you mean by Canonical Cover? Consider the following set F of functional dependencies on schema {A, B, C} :

$A \rightarrow BC$, $B \Rightarrow C$, $A \rightarrow B$, $AB \Rightarrow C$ Compute the Canonical Cover of F.

(c)

(i)

Discuss loss less join decomposition and loss join decomposition.

(ii)

Consider the relation schema R(A, B, C, D, E) into (A, B, C) and (A, D, E).

Show that this decomposition is a loss less decomposition, if the following set of Functional Dependencies $F = \{A \rightarrow CD, CDE, B \rightarrow D, E \rightarrow A\}$. Find out whether the dependency preserved or not.

Attempt **any two** parts of following:

(10x2=20)

(a) What is a Transaction? Explain the properties of the transaction. Explain the States of the transaction with a neat sketch.

(b)

Briefly describe immediate database modification for log based recovery. Explain role of check point in log base.

(c)

What is serializability? What are its types? What is a serializable schedule? Give examples.

Q5.

Attempt **any two** parts of following:

(10x2=20)

(a)

What is locking? Explain two phase locking and its types.

(b)

Explain Timestamp Based Concurrency Control Protocol.

(c)

What is deadlock? Explain necessary conditions for deadlock and methods for handling it.

TCS-404/2120

(2)

PAPER ID-

Subject code

Roll No. to be filled in your Answer Book

Roll No.

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158

Computer Science Engineering,4th Sem

DATABASE MANAGEMENT SYSTEM(TCS/TIT-404)

Time- 3 Hours

NOTE:

Max marks: 100

i.

ii.

All questions are compulsory.

Draw diagrams wherever necessary.

iii. to Give suitable examples wherever necessary.

iv.

All questions carry equal marks.

1. Attempt any **FOUR** parts of **the** following.

dī ban 22, 31

5X4

admin (A) **What** do you understand by the term DBMS explain with a suitable diagram?

Signes

(B) Explain the concept of E-R Diagram with example.

(C) Differentiate between Generalization **and** Specialization.

(D) Discuss at least two problems that occurs because of redundancy in database.

(E) Define DDL and DML in context with DBMS

2. Attempt any **FOUR** parts of the following

5X4

(A) Explain the concept of referential integrity constraints also explain their significance in DBMS.

bosde (B) What do you understand by relational data model and what is its importance justify

your answer?

abs (C) Explain different types of join in SQL giving suitable example in brief.

(D) What do you understand by functional dependency explain. Also differentiate between trivial and non-trivial functional dependency with example.

(E) Explain the Union and Intersection operation in SQL with example.

3. Attempt any **TWO** parts of the following

10X2

(A) What is Normalization? Explain the different **type** of normalization techniques in detail.

(B) For the relation $R(A, B, C, D, E, F, G, H)$ we have following set of functional dependencies $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$. (0-T (1) Write down **the set** of all candidate keys possible for the relation.

(ii) Identify the highest normal form possible for the relation.

(C) Explain in brief dependency preserving decomposition and loss less join decomposition. Also let $R(A, B, C, D)$ be a relation schema with **the** following functional dependencies $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B\}$. If we decompose R into (A, B) , (B, C) and (B, D) state whether the dependency is preserved or not in this decomposition.

4. Attempt any **TWO** parts of the following

10X2

(A) Explain ACID properties in detail with suitable examples. Also explain the importance of these properties in DBMS.

(B) Differentiate between the following:

(i)

(ii)

Concurrent schedule and serial schedule.

Recoverable and irrecoverable schedule.

(C) What do you understand by Conflict serializable schedule and View serializable

schedule **explain** in detail with suitable example.

5. Attempt any **TWO** parts of the following

10 X 2

(A) Explain the significance of locking in database and differentiate between **shared** lock and exclusive lock with suitable example.

(B) What do you understand by concurrency control? **What** is **two** phase locking and its atsitrons types **explain**.

(C) What is deadlock? Explain various deadlock handling techniques.

(TCS/TIT-404)

[Time: 3 Hours]

Odd Semester Examination 2017-18

B Tech III Semester

TCS-301 (CSE) - DISCRETE STRUCTURES

[Maximum Marks: 100]

Attempt all questions. All questions are compulsory.

1. Attempt any four questions. (5×4=20)

- A. For any Set A & B, Prove that $P(A \cap B) = P(A) \cap P(B)$
- B. Explain Pigeonhole principle. If 9 books are to be kept in 4 shelves, show that there must be at least one shelf which contains at least 3 books.
- C. Let $A = \mathbb{Z}^+$, the set of positive integers, and let $R = \{(a, b) \in A \times A \mid a \text{ divides } b\}$. Is R is symmetric or antisymmetric. Also relate symmetric and antisymmetric properties of any relation to the properties of its matrix.
- D. Prove that $A - (B \cup C) = (A - B) \cap (A - C)$.
- E. How many one-one functions are there from a set with m elements to one with n elements?

2. Attempt any four questions. (5×4=20)

- A. Use quantifiers to express the statement "There is a woman who has taken a flight on every airline in the world."
- B. Show that the set of N Natural numbers is a semi group under the operation $x * y = \max(x, y)$.
- C. Construct the truth table for $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow [p \rightarrow r]$
- D. Show that $(p \wedge q) \leftrightarrow (p \vee q)$ is a tautology.
- E. If $(G, *)$ is a group then prove that $(a * b)^{-1} = a^{-1} * b^{-1}$ where $a, b \in G$

3. Attempt any two questions. (10 *2=20)

- A. What is meant by Hasse Diagram? Draw the Hasse diagram of relation R on A where $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (2, 2), (2, 4), (1, 3), (3, 3), (3, 4), (1, 4), (4, 4)\}$
- B. What is complemented lattice and bounded Lattice? Show that lattice is distributive if and only if $(a * b) \vee (b * c) \vee (c * a) = (a \vee b) * (b \vee c) * (c \vee a)$
- C. Is $x * y = x - y$, a binary operation over the set of natural numbers. Is it commutative or associative, Justify your answer for all cases.

4. Attempt any two questions. (10 *2=20)

- A. Differentiate between tautology and contradiction.
- B. Which of the two lattices $(S, \mid, D)$ for $m = 30$ and 45 are complemented? Are these

lattices distributive?

C. Let $A = \{1, 2, 3, 4\}$ and $R = \{(2, 1), (2, 3), (3, 2), (3, 3), (2, 2), (4, 2)\}$. Define R^+ to be the reflexive closure of R and R_s to be the symmetric closure of R . Prove or disprove the symmetric closure of R^+ is the same relation as the transitive closure of R_s .

5. Attempt any two questions. (10 *2=20)

A. Prove that by mathematical induction that $6n+2+72n+1$ is divisible by 43 for each positive integer.

B. Find all the partitions corresponding to a set of four elements. Draw the diagram of the corresponding lattice.

C. In any Boolean algebra show that $a = 0 \Leftrightarrow ab' + db. = b$.

**Sub Code: ECT033 ROLL NO..... ODD SEMESTER EXAMINATION,
2024 – 25**

2nd Year (III Sem) B.Tech.

Digital Electronics

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any two parts of the following. (10x2= 20) a) Convert the following numbers to the indicated bases (i) $(673.124)_8 = ()_2$ (5 marks) (ii) $(153.513)_{10} = ()_8$ (5 marks) b) Simplify the following expression using K-Map Method. (10 marks) (i) $\sum(0,1,2,4,5,6,8,9,12,13,14)$ (ii) $\sum(1,3,7,11,15) + \sum(0,2,5)$ c) Write down the truth table for converting Binary codes to BCD codes, Gray Codes and Excess-3 codes. Also derive the simplified SoP expressions for Binary to BCD converter. (10 marks)
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Draw the block schematic of a one-bit magnitude comparator and explain its operation. (5 marks) (ii) Draw Block diagram of 4:1 Multiplexer and write its truth table. (5 marks) b) Explain the working of a Full Subtractor using the Truth Table and derive the simplified SoP expression for the difference and borrow of a Full Subtractor and draw the logic diagram. (10 marks) c) Design a 4-bit BCD adder circuit. Explain the functioning and draw the logic diagram. (10 marks)
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) Define Finite State Machine. What is the difference between Mealy and Moore machines? (5 marks) (ii) Write down the truth table, excitation table, characteristic equation and logic diagram of JK Flip Flop. (5 marks) b) Design a Universal Shift Register and explain its operation using block diagram. (10 marks) c) Design a BCD/Decade synchronous UP counter using a T Flip Flop. (10 marks)

Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) Classify the different types of programmable logic device. (5 marks) (ii) Distinguish between volatile and non-volatile memory. (5 marks) b) Compare TTL and CMOS logic families on the basis of following: i) Propagation delay ii) Power dissipation iii) Fan-out iv) Basic gate (10 marks) c) How does Programmable logic devices differ from FPGA? Formulate the implementation of the following functions with PLA having three inputs, four product terms, and two output. $f_1(x_1, x_2, x_3) = (3, 5, 6, 7)$ and $f_2(x_1, x_2, x_3) = (0, 2, 4, 7)$. (10 marks)
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Q 5.	Answer any two parts of the following. (10x2= 20) a) Explain the following concepts with respect to VHDL (i) Timing Concept (5 marks) (ii) Gate Delay (5 marks) b) What are the faults in ASIC design? Explain the design flow process suitable for ASIC. (10 marks) c) Enumerate operators, operands and expressions in VHDL with suitable examples. (10 marks)
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Odd Semester Examination 2018-19

B.TECH. (SEMESTER-III)b

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DIGITAL ELECTRONICS AND DESIGN ASPECTS

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Time: 03:00 Hours

Max Marks: 100

ength & won nielaxe balA golfgift XL betepph 6p53 to prichow naloze zus weid **Note:**
Attempt all questions.

1. We' Attempt **any four** questions from the following:

(6)

[4x5=20]

(a)

Convert $(35614)_7$ into decimal.

(b)

Write short note on hamming code.

(c)

A logical express is given as

$$Y = ABC + ABC + ABC + ABC$$

Minimize it using K Map technique.depen

(d)

Perform $(1011)_2 - (0100)_2$ using 1's complement method.

(e)

Minimize the following Boolean expression using K-Map and realize it using NAND gates:

$$Y = [m(1,3,5,6,9,11,13) \& d(7, 15)]$$

(f)

Write short note on De Morgan's theorem.

2.

Attempt **any four** questions from the following:

(a)

Design a full adder circuit.

(b)

Design circuit of a half Subtractor.

(c)

Realize following 3-variable Boolean function using 8:1 MUX :

$$Y = \sum m(1,3,4,6)$$

TEC-302/920

(1)

$$[4 \times 5 = 20]$$

[P.T.O]

3.

(d)

(e)

Implement following Boolean function using 3:8 decoder and external gates. T

$$f(A,B,C) = \sum m(2,4,5,7) \quad 1013enimaxetams \quad bbü$$

Design 4:16 decoder using two 3:8 decoder. 1.8

(f) Write short notes on PAL and PLA. БДНОЯТОВЕ ЛАТНОТО

Attempt **any two** questions from the following: 00:26M YEM

(a)

(b)

JATIDIO

[2x10=20]

BLOH 00:50 T

Draw and explain working of Edge triggered JK flipflop. Also explain how it differs with T flip flops.

With a neat diagram explain operation of a 4 bit left shift register. Also draw its timing diagram.

(c)

Convert JK flipflop into D and SR flipflop.

4.

Attempt **any two** questions from the following:

[2x10=20]

(a)

Discuss in detail about TTL and ECL logic families.

(b)

Describe following characteristics of logic families-

Fan out, Fan in, Figure of Merit, propagation delay and Noise Margin

(c)

Implement below function using CMOS logic

$$Y=A+B+C$$

5.

Attempt **any two** questions from the following:

[2x10=20]

(a)

What is meant by Hazard in a logic network? Discuss different types of Hazards.

(b)

Draw circuit diagram of MOS static RAM cell. Also explain its working.

(c)

(i)

Write short note on DRAM. Explain its advantages and disadvantages.

TEC-302/920

(ii)

How many 256×8 memory chips are required to obtain a 2048×8 memory?.

(2)

Sub Code: BECT 303 ROLL NO..... Back Paper

ODD SEMESTER EXAMINATION, 2023 – 24

B.Tech. 2nd Yr- E&CE/CS&E

DIGITAL ELECTRONICS

Duration: 3:00 hrs Max Marks: 100 *Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.*

Q 1.	Answer any four parts of the following. a) Convert the following in binary: (53) ₁₀ , (45) ₁₀ , (15) ₁₀ , (34) ₁₀ , (31) ₁₀ b) find two's complement of: 01100100, 11011000, 10010010, 01100111, 11111111 c) Construct Hamming code for BCD 0110. Use even parity. d) Prove that $AB + A\overline{B}C + B\overline{A}C = AC + B\overline{A}C$. e) Minimize the following Boolean function $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$ f) Design XOR Gate using NOR Gate.	5x4=20
Q 2.	Answer any four parts of the following. a) Design and implement full adder circuit. b) Design and implement decimal to BCD encoder. c) What is multiplexer? Design 4:1 multiplexer. d) Write a short notes on PAL and PLA. e) Design and implement full Subtractor circuit. f) Explain carry look ahead adder.	5x4=20
Q 3.	Answer any two parts of the following. a) What is a flip-flop? Design and implement J-K flip-flop. b) Convert T flip-flop to S-R flip-flop	10x2=20

	c) Write and explain Ripple Counter.	
Q 4.	Answer any two parts of the following. a) What is the difference between asynchronous and synchronous sequential circuit? Write some applications of asynchronous sequential circuit. b) Explain the operation of CMOS NAND Gate and NOR Gate. c) What is hazard and how to determine hazards in Combinational Circuits? Explain with example.	10x2= 20
Q 5.	Answer any two parts of the following. a) What is ROM? Explain different types of ROM. b) Explain speed of operation, power dissipation, figure of merit, fan in, fan out in detail. c) What is Transistor Transistor Logic (TTL)? Explain operation TTL NAND Gate.	10x2= 20

Sub Code: ECT 033 ROLL NO.....

ODD SEMESTER EXAMINATION, 2023 – 24

IInd yr, B.Tech. – E&CE/CS&E/AI&ML/IT/EE

Digital Electronics

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Find the Binary codes for the Gray codes, (i) 1000 (ii) 1010 b) Write down the basic laws of Boolean Algebra and also write DeMorgan's Theorem. c) Describe 4 digit multiplexer display system. d) Explain Race Around Condition in J-K Flip Flop. e) Explain Tri-state TTL. f) Write a short note on different modeling styles in VHDL	5x4=20
Q 2.	Answer any four parts of the following. a) What are the various classes and types of data objects in VHDL. b) Minimize the following logic function using K-map and implement using logic gates, $Y(A, B, C, D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$ c) Design a Full Adder using Half Adder. d) Explain Master Slave J-K Flip- Flop. e) Explain the working of a two – inputs TTL NAND gate with the help of circuit diagram. f) Explain VLSI design flow.	5x4=20
Q 3.	Answer any two parts of the following. a) Design Gray to Binary Code Converter. b) Implement $Y = \overline{A}BC + A\overline{B}C + AB\overline{C} + \overline{A}\overline{B}\overline{C}$ using multiplexer. c) Explain the components of ASM Charts and describe its salient features.	10x2=20
Q 4.	Answer any two parts of the following. a) Explain interfacing of a TTL gate driving CMOS gates and vice versa. b) Explain Johnson Counter or Twisted Ring Counter. c) Explain Barrel Shifter and ALU.	10x2=20

Q 5.	Answer any two parts of the following. a) Simplify the following four-variable Boolean functions using Quine-McCluskey method, $Y(A,B,C,D) = \sum m(2, 4, 5, 9, 12, 13)$ b) Define the following parameters of logic families and give typical values for a standard TTL gate: (a) Propagation delay (b) Fan-Out (c) Noise Margin (d) Figure of Merit c) Explain Field Programmable Logic Array (FPGA).	10x2= 20
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Sub Code: ECT-033 ROLL NO..... EVEN SEMESTER EXAMINATION, 2024 – 25

II Year (IV Sem) B.Tech.

DIGITAL ELECTRONICS

Duration: 3:00 hrs Max Marks: 100 Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) Find the Gray code equivalent of decimal 13 and the binary equivalent of Gray code number 1111. (ii) Subtract $(-64)_{10}$ from $(+32)_{10}$. Use 2's complement arithmetic. b) Why are NAND and NOR gates called universal gates? Justify your answer with the help of examples. c) Obtain the minimal expression for $F(A, B, C, D) = \sum m(1, 2, 3, 9, 12, 13, 14) + \sum d(0, 7, 10, 15)$ using Quine–McCluskey Tabular Method.
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) Implement a full adder circuit using two half adders and an OR gate. (ii) Implement a Boolean function $F(A, B, C, D) = \sum m(1, 3, 4, 11, 12, 13, 14, 15)$ using a multiplexer. b) What are decoders? Design gate level implementation of 3:8 decoder. c) What is magnitude comparator? Draw the logic gate implementation of 4-bit magnitude comparator.
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) What is a flip-flop? Show the logic implementation of an R-S flip-flop having active HIGH R and S inputs. (ii) Differentiate between level-triggered and edge-triggered flip-flops; b) With the help of a schematic arrangement, explain how a J-K flip-flop can be used as a (a) a D flip-flop and (b) a T flip-flop. c) Draw and explain a 3-bit binary UP/DOWN counter.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) What do you understand by the term logic family? What is the significance of the logic family with reference to digital integrated circuits (ICs)? (ii) Define propagation delay, power dissipation, speed–power product, fan-out and noise margin parameters. b) Explain the Transistor-Transistor Logic (TTL) logic family. Draw and explain circuit operation of Standard TTL NAND gate. c) What are the features of Emitter Coupled Logic (ECL) logic family? With the help of a relevant circuit schematic, briefly describe the operation of ECL OR logic.

Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) Write the VHDL gate level description of 4:1 multiplexer. (ii) Write the VHDL dataflow level description of 4-bit adder. b) Write a VHDL behavioral level description of 8-bit arithmetic logic unit (ALU) using case statement. c) Write and verify an HDL behavioral description of a positive-edge triggered D flip-flop with active low asynchronous preset and clear input.
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